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Sectoral bubble formation and contagion dynamics in China's stock market ☆

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ABSTRACT

This study investigates the drivers of sectoral bubbles and contagion dynamics in China's stock market, addressing an important gap in understanding how bubbles emerge within sectors, propagate across industries, and evolve into market-wide instability. Existing research has largely focused on aggregate market bubbles, paying limited attention to sectoral origins and the time-varying mechanisms through which contagion spreads. Using sector indices from the CSI 300 covering 2011–2023, we employ the modified Backward Sup Augmented Dickey–Fuller (BSADF) method to identify bubble episodes, complementary log–log models with Shapley Additive Explanation (SHAP) values to examine determinants, and time-varying Granger causality tests to trace contagion pathways. We identify three major bubble episodes (2015, 2018, and 2020), with investor behavior emerging as the dominant driver and the Financials and Telecommunications sectors acting as key contagion sources. Importantly, sectoral contagion signals tend to precede market-wide bubbles, offering valuable insights for early-warning systems and targeted policy interventions to mitigate systemic risk.

1. Introduction

Financial markets are periodically marked by phases where asset prices substantially exceed their fundamental valuations, phenomena widely recognized as bubbles (Xiong, 2013). The prevailing definition of bubble involves asset prices that significantly and persistently deviate from their intrinsic values (Kindleberger et al., 2005). These bubbles can precipitate severe economic consequences. During their expansion phase, bubbles often lead to inefficient capital allocation and excessive trading activity. Upon their collapse, they can trigger profound financial crises and extended economic downturns (Xiong, 2013). Hence, there has been a notable increase in scholarly interest in understanding the dynamics of financial bubbles.

Although existing studies have yielded fruitful results regarding the drivers of bubbles (Barberis et al., 2018; Greenwood et al., 2019; Liao et al., 2022), they predominantly focus on the aggregate market while remaining silent on sectoral dynamics. This is a significant omission because stocks within the same sector share common fundamentals and risk exposures, making sectors a natural breeding ground for bubbles (Choi et al., 2015; Milunovich et al., 2019). Sectoral bubbles can therefore act as early warning signals and may even serve as precursors to broader market crises. Understanding the occurrence of such bubbles and their transmission across sectors is crucial for designing more effective early-warning systems and targeted policy responses. In particular, timely regulations could help contain localized bubbles before they escalate into systemic risks that threaten overall financial stability.

Against this backdrop, the primary objective of this study is to investigate the explosive price dynamics from a sectoral perspective, with a focus on the Chinese stock market over the sample period. As the world's second-largest equity market, China presents a unique environment for studying bubble behavior due to several distinguishing features. Firstly, the market is predominantly driven by retail investors, who account for approximately 90% of daily trading volume on the Shanghai Stock Exchange (SSE) (Titman et al., 2022). These investors tend to engage in speculative trading driven by herding and limited financial literacy, contributing to heightened market volatility and a pronounced susceptibility to bubbles (Berger and Turtle, 2015; Bali et al., 2017; Hsiao and Tsai, 2018).[1] Consequently, stock prices may become substantially detached from underlying economic fundamentals. Secondly, stringent restrictions on short selling curtail arbitrage

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[1] As an example of the 2015 stock market bubble, the CSI 300 Index surged from 2300 points in mid-2014 to a peak of 5353 points on June 12, 2015, marking an increase of over 130% in less than a year. During the same period, margin financing soared from RMB 400 billion in mid-2014 to RMB 2.27 trillion by June 2015. The subsequent market crash triggered a liquidity crisis, resulting in severe losses for numerous retail investors. In response, the government introduced "national team" rescue measures to stabilize the market.

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opportunities that could otherwise help correct overvaluations, thereby weakening corrective market forces (Leippold et al., 2022; Xiong and Yu, 2011). Finally, the distinctive structure of the Chinese economic system leads to the sectoral heterogeneity observed in market bubbles. Sectors such as banking, industrial, and utilities are dominated by state-owned enterprises (SOE) and exhibit stable profitability due to rigorous regulatory oversight, making them less susceptible to market whims (Allen et al., 2024). In contrast, sectors like Consumer Discretionary and Healthcare, characterized by higher growth potential, are more influenced by market dynamics. This disparity often results in "structural bubbles" confined to specific sectors of the Chinese stock market (Ji and Zhang, 2024). In light of this, we seek to answer the following questions: When do market- and sector-level bubbles arise in the Chinese stock market? How are investor behavior, sectoral fundamentals, and macroeconomic conditions associated with the emergence of these bubbles? Do sector-specific bubbles have the potential to spread to the broader market?

To answer these questions, our empirical strategy unfolds in three stages. We begin by identifying explosive episodes in the CSI 300 Index and its ten Level-1 sector indices by employing the modified Backward Sup Augmented Dickey–Fuller (mBSADF) test by Wang et al. (2023).[2] These results establish when and where bubbles arise, thereby identifying both market-wide bubbles and sector-specific origins. Using the identified bubble episodes, we then examine the determinants of bubble occurrence by estimating complementary log–log (clog–log) models and Shapley Additive Explanation (SHAP) values, which allow us to quantify both the marginal effects and the relative importance of investor behavior, sectoral fundamentals, and macroeconomic conditions. Finally, we explore how sectoral bubbles can spill over into other sectors and the aggregate market with the time-varying Granger causality test by Shi et al. (2018). By integrating these analyses in a sequential framework, we provide a comprehensive picture of bubble occurrence, its underlying determinants, and the transmission pathways through which sectoral exuberance may evolve into broader market instability.

This study provides a systematic empirical analysis of explosive behavior at the sectoral level in the Chinese stock market through a unified framework that identifies bubble episodes, investigates their underlying determinants, and traces their transmission across market segments. We begin by documenting and characterizing sectoral bubble phenomena. Beyond revisiting the well-documented 2015 market-wide crash, our analysis detects two previously underexplored episodes of explosive behavior that are confined to individual sectors, thus conceptualizing them as "structural bubbles" (Ji and Zhang, 2024). To understand the factors associated with these sectoral bubbles, we examine investor behavior, sector fundamentals, and macroeconomic conditions at the sectoral level, departing from prior studies which focus on the impact of an individual or a handful of factors on the occurrence of market-wide bubbles. Methodologically, we complement the existing literature by employing a complementary log–log (clog–log) model to accommodate "rare-event bias" inherent in bubble studies, thereby yielding more suitable estimates for asymmetric binary outcomes.[3] Finally, we dynamically trace the contagion pathways of sectoral bubbles. Extending beyond static frameworks, we introduce a time-varying perspective to analyze how bubbles propagate not only across sectors but also from specific sectors to the aggregate market. Our application of time-varying Granger causality tests captures the evolution of contagion direction and intensity, uncovering sectoral "leaders" in bubble transmission and offering novel insights into how localized imbalances escalate into systemic risk.

The findings of this study can be summarized as follows. First, the Chinese stock market has experienced sustained explosive behaviors in 2015, 2018, and 2020 over the sample period. The 2015 episode is characterized by a widespread surge in prices across all sectors. In contrast, the explosive price behaviors in 2018 and 2020 are confined to specific sectors, thus being identified as "structural bubbles". Second, investor behaviors, encompassing speculative activities, liquidity, and market sentiment, emerge as the most critical factor influencing the probability of bubble occurrences. High-growth sectors, such as Consumer Discretionary, Health Care, and Materials, are most strongly associated with these factors. These sectors are more susceptible to irrational market behaviors compared to sectors such as Financials, Energy, and Utilities, which are less impacted due to stringent regulatory oversight and inherent stability. Furthermore, we reveal that the Financials and Telecommunication sectors tend to act as unidirectional sources of bubble contagion to other industries. Consistent with their central positions in capital flows, investor sentiment, and policy exposure, bubbles in these sectors appear more likely to amplify and spill over to the broader market. In contrast, sectors subject to strict regulatory oversight, such as Industrials, Utilities, and Health Care, exhibit lower contagion potential. Notably, we find that contagion signals often precede the emergence of market-wide bubbles, underscoring their value as early-warning indicators. These empirical patterns suggest a plausible pathway of bubble evolution in the Chinese stock market, in which macro-financial conditions and investor behavior are associated with sectoral bubbles that may, in turn, contribute to broader market instability. This highlights an opportunity to anticipate and mitigate systemic market risks based on sector-specific dynamics. Our main findings are substantiated through a series of robustness checks.

The rest of this paper is organized as follows. Section 2 reviews the existing relevant literature. Section 3 describes the employed methods and the dataset. Section 4 reports the empirical results and discusses the practical implications. Section 5 concludes this paper.

2. Literature review and hypothesis development

2.1. Asset price bubbles

The presence of asset price bubbles has been well documented in existing literature, especially in emerging markets such as China (Giglio et al., 2016; Basse et al., 2021; Hudepohl et al., 2021). Widely used bubble detection techniques fall into two primary strands. The first strand uses the Log-Periodic Power Law Singularity (LPPLS) model, which has been applied to detect multiple major bubble episodes in China's equity market, including the periods of 2005 – 2007, 2008 – 2009, and the market boom and bust of 2014 – 2015 (Sornette et al., 2015). The second strand of research relies on the Backward Supremum ADF (BSADF) (Wang et al., 2021; Zhao et al., 2021; Yu and Li, 2023).

However, market-wide evidence can mask substantial cross-sector heterogeneity in bubble dynamics. Stocks within the same sector share common fundamentals, business-cycle exposure, regulatory environments, and investor clienteles, making sectors natural units in which speculative dynamics may emerge (Milunovich et al., 2019). Moreover, sectors may differ in their susceptibility to explosive dynamics. For

[2] The CSI 300 Index, which includes the 300 largest and most liquid stocks from the Shanghai and Shenzhen stock exchanges, along with its sector indices, serves as a comprehensive representation of both overall and sector-specific market dynamics in China. The chosen indices are widely recognized proxies for monitoring the broader market and its distinct sectors, capturing the nuanced volatility and sector-specific bubbles effectively.

[3] For example, Phillips et al. (2015a) examined the S&P 500 index from January 1871 to December 2010 and identified only five bubble periods out of 1680 monthly observations, accounting for just 6.5% of the dataset. This results in an asymmetric distribution of the binary bubble variable, representing the occurrence of bubbles, which introduces "rare event bias" when traditional probit and logit models are used (King and Zeng, 2001). The complementary log–log (clog–log) model addresses this limitation by accommodating the asymmetry in the distribution, thereby yielding more reliable estimates.

example, growth-oriented or narrative-sensitive sectors, compared with mature or highly regulated sectors, are more exposed to attention-driven demand and optimistic growth stories (Pástor and Veronesi, 2009; Shiller, 2020). Existing sector-level studies provide important but incomplete evidence on this issue. In developed markets, Anderson et al. (2010) use a regime-switching approach to investigate bubbles in sector indices of the S&P 500, while Greenwood et al. (2019) examine sector-level bubble episodes across the US and international markets. These studies show that bubble dynamics are not limited to aggregate indices and may be concentrated in particular industries. In emerging-market and broader sectoral settings, Narayan et al. (2013) identify sectors such as electricity, energy, financial, and banking as particularly prone to bubble formation, emphasizing sectoral susceptibility. Similarly, Yang and Ferrer (2023) document periods of explosiveness across equity sectors and the aggregate equity market.

Taken together, this literature suggests that sectoral bubbles are economically meaningful phenomena rather than mechanical decompositions of market-wide bubbles. Despite these extensive studies, gaps remain in our understanding of the precise triggers and consequences of sectoral bubbles, which our research aims to address. Given the above reasoning, we propose the first hypothesis:

Hypothesis 1. Bubble episodes in China's stock market exhibit substantial sectoral heterogeneity in their timing, duration, and intensity.

2.2. Determinants of sectoral bubble formation

Based on prior literature, determinants of bubble formation can be broadly classified into three primary categories: investor behavior, sectoral fundamentals, and macroeconomic conditions. Investor behavior may drive bubble formation through speculative trading, liquidity, and sentiment. The process typically begins with speculative trading, where investors purchase assets primarily for anticipated future price gains rather than fundamental value (Bayer et al., 2021). As early speculators realize substantial profits, these gains attract additional speculative capital, creating a positive feedback loop in which rising prices generate further buying interest (Shiller, 2000). Liquidity amplifies this cycle by reducing trading frictions and enabling speculative demand to be quickly translated into price pressure (Liao et al., 2022; DeFusco et al., 2022). This process is also consistent with gradual information diffusion and trend-chasing behavior, in which investors extrapolate recent price increases and continue to buy assets with strong past returns (Hong and Stein, 1999).

Investor sentiment provides the psychological foundation that sustains these dynamics beyond what fundamental valuations would justify. Overconfidence, limited attention, and extrapolative expectations may cause investors to underestimate downside risk and overestimate their ability to time the market (Greenwood and Shleifer, 2014; Liu et al., 2015). These conditions are especially relevant in China's A-share market, where retail investors account for a large share of trading activity (H. Liu et al., 2023), short-selling remains structurally constrained (Deng et al., 2020), daily price limits may delay price adjustment (Chen et al., 2019), and policy announcements often serve as focal points for coordinated sentiment shifts (Zhou, 2018). In such an environment, limits to arbitrage allow sentiment-driven mispricing to persist and compound over time.

At the sectoral level, these behavioral mechanisms may be stronger because investor attention and speculative narratives are unevenly allocated across industries (Shiller, 2020). Investors tend to focus on sectors that are salient, recently visible, or tied to compelling growth stories (Kumar, 2009). As attention and trading become concentrated in a sector, sentiment and speculative demand can jointly push sector prices away from fundamentals, generating localized bubbles before these dynamics appear in the aggregate market (Zhou and Zheng, 2026).

The second strand of literature on bubble formation focuses on sectoral fundamentals. Profitability metrics may affect bubble formation through their effects on growth expectations, risk perceptions, and valuation multiples (Yan and Zheng, 2017; Hou et al., 2015). Aggressive asset growth may increase bubble susceptibility by signaling expansion opportunities that investors systematically overvalue, even though rapid growth may later be associated with weaker future returns as operational difficulties emerge (Gray and Johnson, 2011). Moreover, institutional ownership may influence bubble susceptibility by affecting information quality and trading dynamics. Higher institutional ownership typically moderates price inflation through improved monitoring, information processing, and price discovery (Andreou et al., 2016; Choi et al., 2015).

The third strand emphasizes macroeconomic conditions, which create the structural environment that facilitates or constrains bubble development. Macroeconomic factors operate through their impact on expected cash flows, discount rates, liquidity conditions, and investor risk tolerance (Basse et al., 2021). GDP growth enhances corporate earnings prospects and may create upward pressure on stock valuations, while inflation affects real cash flows and operating costs across sectors (Allen et al., 2024). Economic policy uncertainty may raise risk premiums and dampen asset prices, but it can also contribute to bubble formation when investors underestimate policy risks or search for higher returns in uncertain environments (Phan et al., 2018; Cheng et al., 2021; Chen et al., 2023). Similarly, accommodative monetary policy reduces discount rates and increases liquidity, creating conditions under which investors may bid up asset prices beyond fundamental values.

In sum, investor behavior, sectoral fundamentals, and macroeconomic conditions provide three complementary perspectives for explaining sectoral bubble formation. Investor behavior creates speculative momentum and sentiment-driven mispricing, sectoral fundamentals determine industry-specific vulnerabilities, and macroeconomic conditions shape the broader valuation and liquidity environment. However, existing research has predominantly examined these factors in isolation or focused on market-wide episodes. What remains underexplored is how these factors are jointly associated with bubble occurrence at the sectoral level and which group of factors has greater explanatory relevance. We therefore propose the second hypothesis:

Hypothesis 2. Investor behavior, sectoral fundamentals, and macroeconomic conditions are jointly associated with the likelihood of sectoral bubble formation.

2.3. Sectoral contagion and market-wide instability

Historically, asset bubbles were often confined to individual sectors. However, bubbles may propagate across industries and markets, thereby affecting broader economic stability (Tian et al., 2023). The contagion effect of financial bubbles is well documented across various asset classes, including cryptocurrencies (Enoksen et al., 2020; Chen et al., 2024), oil, gold, stock markets (Zhao et al., 2021; Bei et al., 2024), and housing markets (Gomez-Gonzalez et al., 2018). This literature shows that bubbles are not isolated pricing anomalies, but dynamic processes that can spread across markets through financial and behavioral linkages.

While bubble contagion across asset classes is well documented, dynamics within stock market sectors remain underexplored. On the economic side, industries may be linked through production networks, credit exposure, common macroeconomic shocks, and policy expectations (Acemoglu et al., 2012; Nguyen and Waters, 2022; R. Huang et al., 2025). On the investor side, bubbles may spread when investors rotate capital from a booming sector to related industries (Z. Huang et al., 2025), extrapolate successful sectoral narratives to comparable sectors (Shiller, 2020), or rebalance portfolios in response to changing risk perceptions (Forbes and Rigobon, 2002). These channels imply that sectoral bubbles may generate spillovers before aggregate market indices fully reflect broader market exuberance.

Although prior studies have examined inter-sectoral bubble transmission (Anderson et al., 2010; Yang and Ferrer, 2023), existing evidence remains limited in two respects. First, much of the literature focuses on whether contagion exists, while paying less attention to the timing and direction of transmission. Second, static frameworks may overlook the fact that contagion channels can strengthen, weaken, or reverse across different bubble episodes. Understanding this temporal dimension is crucial for identifying leading and receiving sectors and assessing whether sectoral contagion signals can provide early warnings of broader market instability.

Hypothesis 3. Sectoral bubbles can transmit across industries and to the aggregate market through time-varying contagion channels. Such contagion dynamics may provide early signals of broader market instability.

3. Methods and data

3.1. Modified backward sup augmented Dickey–Fuller test

We begin by identifying bubble episodes at both the aggregate market and sectoral levels. The Backward Sup Augmented Dickey–Fuller (BSADF) of Phillips et al. (2015a,b) is widely used for real-time bubble detection, but it can have limited power for short-lived explosive episodes and may generate spurious signals (Wang et al., 2023). To address these issues, we adopt the modified BSADF (mBSADF) procedure proposed by Wang et al. (2023), which augments the BSADF statistic with a modification term to improve sensitivity to short-duration explosive episodes and reducing false positives.[4]

Consider a price series $\{y_t\}_{t=1}^{T}$ generated by

$$y_t = dT^{-\eta} + \rho y_{t-1} + u_t, t = 1, 2, \dots, T, \tag{1}$$

where $dT^{-\eta}$ is an asymptotically negligible drift, with d a constant, T the sample size, and $\eta > 1/2$ a localizing parameter. When $\eta > 1/2$, the drift component vanishes asymptotically and y_t behaves like a random walk under the null hypothesis.[5] Under the null hypothesis $H_0 : \rho = 1$, y_t follows a unit root process, whereas under the alternative $H_1 : \rho > 1$, it exhibits explosive behavior.

The mBSADF statistic is constructed recursively through a two-step procedure. For a given fraction endpoint $f \in [f_0, 1]$ an ADF regression is first estimated over a subsample starting at fraction f_1 and ending at f:

$$\Delta y_t = \hat{\alpha}_{f_1,f} + \hat{\beta}_{f_1,f} y_{t-1} + \sum_{j=1}^{p} \hat{\theta}^{j}_{f_1,f} \Delta y_{t-j} + u_t, t = \lfloor Tf_1 \rfloor, \dots, \lfloor Tf \rfloor, \tag{2}$$

where Δy_t denotes the first difference of y_t, the lag order p controls potential autocorrelation and u_t is an i.i.d. error term with mean zero and finite variance. The corresponding ADF statistic is defined as

$$ADF_{f_1,f} = \hat{\beta}_{f_1,f} / se(\hat{\beta}_{f_1,f}).$$

Next, the starting point f_1 is rolled backward from $f - f_0$ to 0, generating a sequence of ADF statistics $\{ADF_{f_1,f}\}_{f_1 \in [0, f-f_0]}$. The mBSADF statistic at endpoint f is then defined as:

$$mBSADF_f(f_0) = \sup_{f_1 \in [0, f-f_0]} ADF_{f_1,f} + \frac{c}{K} \sum_{t=\lfloor Tf \rfloor - K+1}^{\lfloor Tf \rfloor} m_t, f \in [f_0, 1], \tag{3}$$

where f_0 is the minimum window size, $m_t = \Delta y_t / \sigma_{\Delta y_t}$, and $\sigma_{\Delta y_t}$ is the standard deviation of $\{\Delta y_t\}$. The second term introduces a K-period moving average modification, scaled by a positive constant c. By rolling the endpoint f forward over the sample, a sequence of mBSADF statistics is obtained. Bubble origination (f_o) and termination (f_c) dates are identified by comparing $mBSADF_f(f_0)$ with its right-tailed critical value:

$$\hat{f}_o = \inf_{f \in [f_0, 1]} \{ f : mBSADF_f(f_0) > mscv_f \},$$
$$\hat{f}_c = \inf_{f \in [\hat{f}_o + L_b, 1]} \{ f : mBSADF_f(f_0) < mscv_f \} \tag{4}$$

where L_b is the minimum bubble duration required to prevent misclassifying short-lived irregular fluctuations as bubbles (Phillips et al., 2015a), and $mscv_f$ denotes the 95% right-tailed critical value of $mBSADF_f(f_0)$. Accordingly, a bubble is dated from the first observation at which the mBSADF statistic exceeds its critical value until the first subsequent observation at which it falls below the critical value.

3.2. Complementary log–log model

After detecting bubble periods, we next examine their occurrence drivers. Prior studies typically apply probit or logit models (Enoksen et al., 2020; Chang, 2024). However, these methods may suffer from “rare event bias”, given that bubble episodes tend to be sparse over the full sample. To mitigate this issue, we adopt the complementary log–log (clog–log) model, which is particularly suited for rare-event settings. Compared with probit and logit, the clog–log model could capture the asymmetries in outcome probabilities and reduce estimation bias. Recent applications in financial contexts further highlight its advantages for modeling rare-event phenomena (Scheubel et al., 2025; Batini and Durand, 2021; Palandökenlier and Bal, 2025).

We construct time-series clog–log regression models for both the CSI 300 and sector indices, to capture market-wide bubble determinants as well as heterogeneity across sectors. The clog–log specification is given as:

$$\begin{aligned} &log(-log(1 - P(Bub_t = 1))) \\ &= \beta_1 ATR_t + \beta_2 QS_t + \beta_3 Sent_t^{\perp} + \beta_4 ATR_t * QS_t + \beta_5 ATR_t * Sent_t^{\perp} + \\ &\quad \beta_6 ROE_t + \beta_7 Insti_t + \beta_8 Asset_t + \beta_9 EPU_t \\ &\quad + \beta_{10} rGDP_t + \beta_{11} dShibor_t + e_t, \end{aligned} \tag{5}$$

where the binary dependent variable of each index is defined as $Bub_t = I(mBSADF_f(f_0) > mscv_f)$, $t := \lceil Tf \rceil$. Eq. (5) includes the three categories of potential determinants, with detailed definitions provided in Table 2. To account for the potential correlation between speculative activity and sentiment, we orthogonalize the sentiment with respect to abnormal turnover rate, thus the residual series ($Sent^{\perp}$) is used. In addition, interaction terms between ATR and both QS and $Sent^{\perp}$ are included to capture possible interactive effects.

To assess the relative importance of these potential drivers, we specifically compute the average marginal effect (AME) of each factor, which measures its average impact on the probability of bubble occurrence. The AME for variable x_i is calculated as

$$AME_i = \frac{1}{T} \sum_{t=1}^{T} \beta_i e^{X_t' \beta - e^{X_t' \beta}}, \tag{6}$$

where X is the vector of independent variables as defined in Eq. (5).

3.3. Time-varying Granger causality test

Another important approach to understanding stock market bubble dynamics is to examine how price movements are transmitted across indices. To this end, we employ the time-varying Granger causality (TVGC) test proposed by Shi et al. (2018) to investigate dynamic causal relationships across sectors and from sectors to the aggregate market,

[4] A detailed comparison between the BSADF and mBSADF procedures is provided in Appendix A.

[5] According to Phillips et al. (2014), when $0 \le \eta \le 1/2$, the standardized series $T^{-1/2} y_t$ converges to a Brownian motion with drift.

proxied by the CSI 300 index. The TVGC framework is well-suited to capture time-varying dependence structures in financial markets and has been widely applied in recent empirical studies (Maghyereh and Abdoh, 2022; Gharib et al., 2021).

For two series y_{1t} and y_{2t}, we consider the following unrestricted $VAR(p)$ model:

$$Y_t = \Pi X_t + \epsilon_t, t = 1, \dots, T, \tag{7}$$

where $Y_t = (y_{1t}, y_{2t})'$, $X_t = (1, Y'_{t-1}, \dots, Y'_{t-p})$ is the regressor vector including an intercept and p lags of Y_t. The coefficient matrix Π has dimension $2\times(2p+1)$, which can be partitioned as $[\Psi_0, \Psi_1, \dots, \Psi_p]$, with $\Psi_0 \in \mathbb{R}^{2\times1}$ and $\Psi_j \in \mathbb{R}^{2\times2}$ for $j = 1, \dots, p$. Denote $\hat{\Pi}$ as the OLS estimator of Π and define the residuals covariance matrix as $\hat{\Omega} = T^{-1}\sum_{t=1}^{T} \hat{\epsilon}_t\hat{\epsilon}'_t$ with $\hat{\epsilon} = Y_t - \hat{\Pi}X_t$, and $X' = [X_1, \dots, X_T]$ denote the regressor matrix so that $\mathbf{X'X} = \sum_{t=1}^{T} X_t X'_t$. To test the null hypothesis H_0: $y_{2t} \nrightarrow y_{1t}$, i.e., y_{2t} is not a Granger cause for y_{1t}, the Wald test statistic is calculated as:

$$\mathcal{W} = [\mathbf{R}\,\mathrm{vec}(\hat{\Pi})]' \left[\mathbf{R}\left(\hat{\Omega}\otimes\left(\mathbf{X'X}\right)^{-1}\right)\mathbf{R}'\right]^{-1} [\mathbf{R}\,\mathrm{vec}(\hat{\Pi})], \tag{8}$$

where $\mathrm{vec}(\hat{\Pi})$ denotes the $2(2p+1)\times 1$ vector obtained by stacking the coefficients of $\hat{\Pi}$, $\otimes$ is the Kronecker product. The matrix $\mathbf{R}$ is a $p \times 2(2p+1)$ selection matrix that imposes H_0 by selecting the p coefficients on the lagged values of y_{2t} in the y_{1t} equation. Employing the recursive evolving algorithm of Shi et al. (2018), the time-varying Granger causality statistic at the fraction f (with $t = \lceil Tf \rceil$) is defined as:

$$\mathcal{SW}_f(f_0) = \sup_{f_1\in[0,f-f_0]} \mathcal{W}_f(f_0), \tag{9}$$

where $\mathcal{W}_f(f_0)$ is the Wald statistic computed on the subsample $[f_1, f]$, and f_0 is the minimum fractional window size. $\mathcal{SW}_f(f_0)$ is the supremum of Wald statistics as the starting point f_1 moves from 0 to $f - f_0$. Let f_o and f_c denote the origination and termination dates of a causal episode. They are dated by comparing $\mathcal{SW}_f(f_0)$ with its critical value scv:

$$\hat{f}_o = \inf_{f\in[f_0,1]} \{f : \mathcal{SW}_f(f_0) > scv\}, \hat{f}_c = \inf_{f\in[\hat{f}_o,1]} \{f : \mathcal{SW}_f(f_0) < scv\}. \tag{10}$$

Accordingly, a significant Granger-causal segment is identified when $\mathcal{SW}_f(f_0)$ exceeds scv. We then assess bubble transmission by examining whether these causal segments overlap with the bubble periods dated by the mBSADF procedure.

Finally, both the mBSADF and TVGC procedures rely on recursive evolving estimation and can be somewhat sensitive to the choice of sample length T and minimum window size f_0. In particular, for empirically large T, the sequential re-estimation inherent in recursive evolving procedures may impose a substantial computational burden. These features are common to recursive approaches and are worth taking into account when applying these methods.

3.4. Data

3.4.1. The CSI 300 and sector indices

This paper analyzes the Chinese stock market using the CSI 300 index along with its 10 level-1 sector indices: Energy, Materials, Industrials, Consumer Discretionary, Consumer Staples, Health Care, Financials, Information Technology, Telecommunication, and Utilities. The sample spans 630 weekly observations from January 7, 2011, to December 29, 2023.[6] This period covers significant events impacting Chinese capital markets, including the 2015 stock market crash and the global ramifications of the COVID-19 pandemic in 2020, providing a representative sample that spans major domestic and global market disruptions.

[6] Data are sourced from https://data.csmar.com/.

Table 1 reports the descriptive statistics for the sample data. The CSI 300 reaches its trough of 2122.84 points in March 2014 and peaks at 5778.84 points in February 2021, reflecting an overall increase of 8.35%. At the sectoral level, with the exception of Energy and Materials, most indices reach their peaks in June 2015 or February 2021, corresponding to the market booms of 2014–2015 and the COVID-19 pandemic. By contrast, sectoral troughs are closely associated with major macroeconomic shocks. The Energy index, for instance, reached its lowest point in June 2020, which was driven by the collapse in global demand during the pandemic (Huang and Liu, 2021). Financials and several other indices record their troughs in late 2012 under the spillover effects of the European sovereign debt crisis (Hsiao and Morley, 2022), and other sectors follow with downturns extending into mid-2014.

In terms of growth, the Energy, Materials, and Industrials experience declines exceeding 30%, whereas most other sectors display growth rates comparable to or even surpassing that of the CSI 300 index. Notably, Consumer Staples and Health Care exhibit the highest cumulative returns (275.36% and 43.97%) and elevated volatility (10,216.88 and 3530.17), reflecting steady demand supported by consumption growth, demographic trends, and healthcare reforms. In contrast, the Financials and Industrials sectors, holding the largest market weights (35.84% and 13.92%), displayed volatility in line with the overall market, reflecting structural stability and the influence of regulatory policies. These heterogeneous patterns in growth, volatility, and extremes underscore the importance of conducting sector-level analyses of bubble dynamics rather than relying solely on market-wide aggregates.

3.4.2. Influencing factors of bubble occurrence

Following the existing literature, we examine three categories of potential drivers of bubble occurrence (proxied by the binary variable *Bub*): investor behavior, sectoral fundamentals, and macroeconomic conditions. The construction and definitions of these variables are detailed in Table 2.[7] Sector-specific firm-level variables (ATR, QS, ROE, $Insti$, $Asset$) are aggregated to the sector level using market capitalization-weighted averages. To ensure consistency in frequency, the quarterly $rGDP$ and monthly EPU are converted to weekly series by assigning each observation to all corresponding weeks. The data set encompasses the CSI 300 Index and 10 primary sector indices over 580 weeks from January 6, 2012, to December 29, 2023, totaling 6380 observations.[8]

Table 3 presents the descriptive statistics of the considered variables. According to the table, the mean value of ATR is 1.026. In the absence of heightened speculative activity, ATR is expected to remain close to unity. Thus, the average level suggests that speculation is not particularly pronounced. Nevertheless, the substantial variation in ATR, ranging from a minimum of 0.192 to a maximum of 4.380, underscores significant fluctuations in speculative behaviors. The mean values of QS and $Sent$ are −0.118 and 0.010, respectively, which are close to their median values. In terms of sectoral fundamentals, the average ROE and $Asset$ stand at 8.7% and 3.6%, respectively. These values indicate stable profitability and moderate expansion in asset bases among listed Chinese firms over the sample period. The institutional ownership ratio ($Insti$) has an average of 70.10%, a high

[7] Data for $Sent$, EPU, $rGDP$, and $Shibor$ are sourced from Bloomberg, https://www.policyuncertainty.com/, China's National Bureau of Statistics https://www.stats.gov.cn/english/, and https://www.shibor.org/shibor/index.html, respectively. The original data series for the sector-specific firm characteristics are sourced from https://data.csmar.com/.

[8] The time span of the stock index series is from January 7, 2011, to December 29, 2023, comprising a total of 630 weeks. However, due to the mBSADF method's minimum window size requirement of $\lceil Tf_0 \rceil = 51$ weeks, the resulting statistics and bubble variable series generated in Eq. (4) correspond to the period from January 6, 2012, to December 29, 2023, encompassing 580 weeks.

Table 1
Descriptive statistics of the CSI 300 and sector indices.

	Mean	Min	Date (min)	Max	Date (max)	Std. Dev.	Weight	Growth
CSI 300	3519.29	2122.84	2014/03	5778.84	2021/02	844.12	/	8.34%
Energy	2022.73	1188.12	2020/06	4176.23	2011/04	643.73	9.54%	−45.78%
Materials	2398.32	1405.54	2014/06	4228.37	2021/09	594.25	6.16%	−31.17%
Industrials	2413.21	1444.08	2014/06	5100.74	2015/06	581.06	13.92%	−33.58%
Consumer Discretionary	5077.93	2597.91	2012/08	9806.50	2021/02	1517.44	7.74%	31.55%
Consumer Staples	15250.06	4884.16	2014/06	42816.80	2021/02	10216.88	10.70%	275.36%
Health Care	8949.66	3976.75	2012/01	19192.20	2021/02	3530.17	5.30%	43.97%
Financials	5228.05	3088.10	2012/09	7715.80	2015/06	1222.00	35.84%	21.30%
Information Technology	2007.39	989.83	2012/11	3768.73	2015/06	561.35	6.61%	3.17%
Telecommunication	2517.86	1290.57	2012/11	4860.96	2015/06	555.38	1.43%	−7.82%
Utilities	1799.06	1120.15	2014/01	3409.18	2015/06	369.67	2.75%	34.30%

Note: This table reports descriptive statistics for the CSI 300 and 10 sector indices based on 630 weekly observations from January 7, 2011, to December 29, 2023. Date (min) and Date (max) indicate when each index reached its minimum and maximum. Weight is the annual average of each sector's year-end capitalization as a share of total CSI 300 market capitalization. Growth denotes the total price increase of each index over the sample period.

Table 2
Summary of variables.

Symbol	Variable	Proxy	Definition
Panel A: Investor Behaviors			
ATR	Abnormal Turnover Rate	Speculation	Ratio of the firm's average daily turnover over the past 20 days to that over the past 250 days (Liu et al., 2019).
QS	Quoted Spread	Liquidity	Ratio of the difference between selling and buying prices to the median of these two prices (Le and Gregoriou, 2020).
$Sent$	Investor Sentiment	Sentiment	J.P. Morgan China A-shares Sentiment Index, which combines technical indicators, capital flows and investor confidence.
Panel B: Sectoral Fundamentals			
ROE	Return on Equity	Profitability	Net profit divided by stockholders' equity, measuring firm profitability.
$Insti$	Institutional Ownership	Institutional Participation	Proportion of shares held by institutional investors, reflecting institutional involvement.
$Asset$	Asset Growth Rate	Growth	Change in total assets over a period divided by total assets at the beginning of the period.
Panel C: Macroeconomic Factors			
EPU	Economic Policy Uncertainty	Economic Policy	China's Economic Policy Uncertainty Index (Baker et al., 2016).
$rGDP$	GDP Growth Rate	Economic Growth	Quarterly GDP growth rate, CPI-deflated to adjust for inflation.
$dShibor$	Change in SHIBOR	Monetary Policy	Weekly changes in Shanghai Interbank Offered Rate, CPI-deflated to adjust for inflation.

level of participation by institutional investors in the equity market. For macroeconomic indicators, the economic policy uncertainty index (EPU) fluctuates between 98.36 and 234.52, with a standard deviation of 28.49, reflecting substantial variations in policy-related uncertainty over time. The average quarterly growth rate of $rGDP$ is 1.57%, suggesting a stable pace of economic expansion during the sample period. Meanwhile, the mean change in the interbank offered rate ($dShibor$) is −0.004%, implying minimal directional shifts and relatively low short-term volatility in interbank interest rates. Furthermore, panel unit root tests, namely the Im–Pesaran–Shin (IPS) test (Im et al., 2003) and the Levin–Lin–Chu (LLC) test (Levin et al., 2002), confirm the stationarity of all series, ensuring their appropriateness for the inclusion of the regression models.

4. Empirical results

4.1. Bubble detection results

This section presents our empirical identification of explosive price behaviors in the Chinese stock market using the modified backward sup ADF (mBSADF) test. The analysis is conducted for both the CSI 300 and 10 sectoral indices, with several parameter specifications tailored to our sample. Specifically, we employ a minimum window of $\lceil Tf_0 \rceil = \lceil 0.2\sqrt{T} \rceil = 51$ observations for our 630-week dataset.[9] As standard, the bubble duration is restricted to a minimum of 3 months. The lag length in the ADF regression (Eq. (2)) is selected by the Bayesian information criterion (BIC), with a maximum lag order of 6 to account for serial correlation while maintaining parsimony. Finally, the finite-sample critical values are derived via Monte Carlo simulation with 2000 repetitions, following Phillips et al. (2015a) and Wang et al. (2023).

Our analysis identifies three distinct episodes of explosive behavior across both market and sector levels during our sample period: 2014–15, 2017–18, and 2020–21. Fig. 1 visually highlights the detected bubble periods, marked in gray, across the sample timeline. For clarity and comparison, these results are summarized in Table 4. The results reveal substantial cross-sector heterogeneity in the timing and duration of bubble episodes, providing direct evidence for Hypothesis 1.

The well-documented 2014–15 episode represents a synchronized market-wide bubble affecting nearly all sectors, driven by policy stimulus, monetary easing, and retail speculation (Sornette et al., 2015; Wang et al., 2021; Yang and Ferrer, 2023). In contrast, the 2017–2018 and 2020–2021 episodes exhibit distinct sector-specific bubble patterns. These bubbles are not pervasive across the entire market but are instead confined to several specific sectors, such as Materials, Consumer Discretionary, Consumer Staples, and Health Care. Sectors like Energy, Financials, and Utilities remain relatively stable throughout. Importantly, several sector-level bubbles precede aggregate market bubbles by several months, suggesting the possibility that sectoral exuberance could serve as an early warning signal of broader market instability. These findings underscore the importance of moving beyond the aggregate index to analyze sectoral heterogeneity in bubble formation.

We also apply the traditional BSADF, and the results are reported in Table B.1. The BSADF confirms the 2014–15 market bubble, partially

[9] Robust bubble estimates can be obtained using $f_0 = 0.01 + 1.8/\sqrt{T}$ as suggested by Phillips et al. (2015a) or using $f_0 = 0.15/\sqrt{T}$. The results are available upon request.

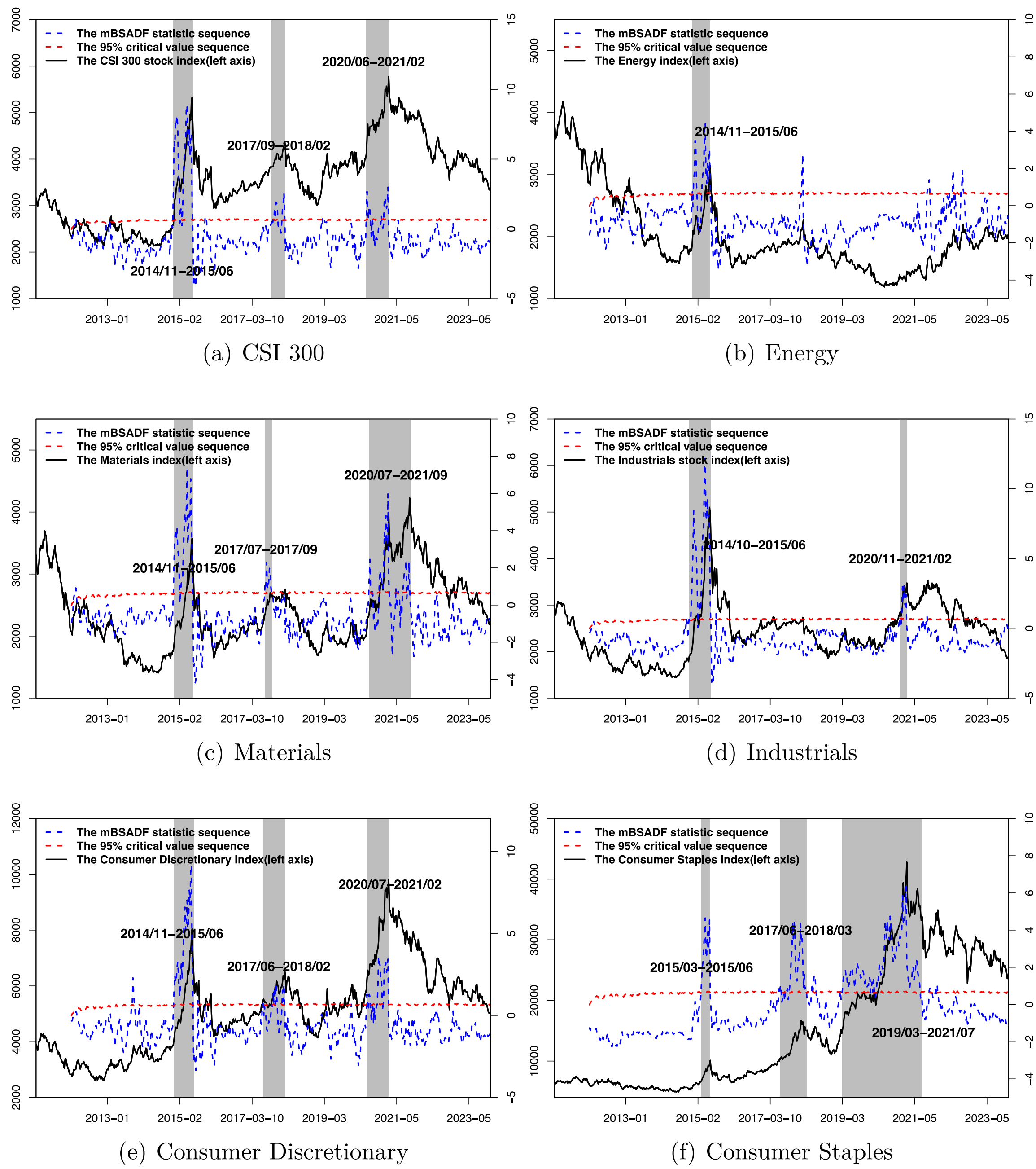
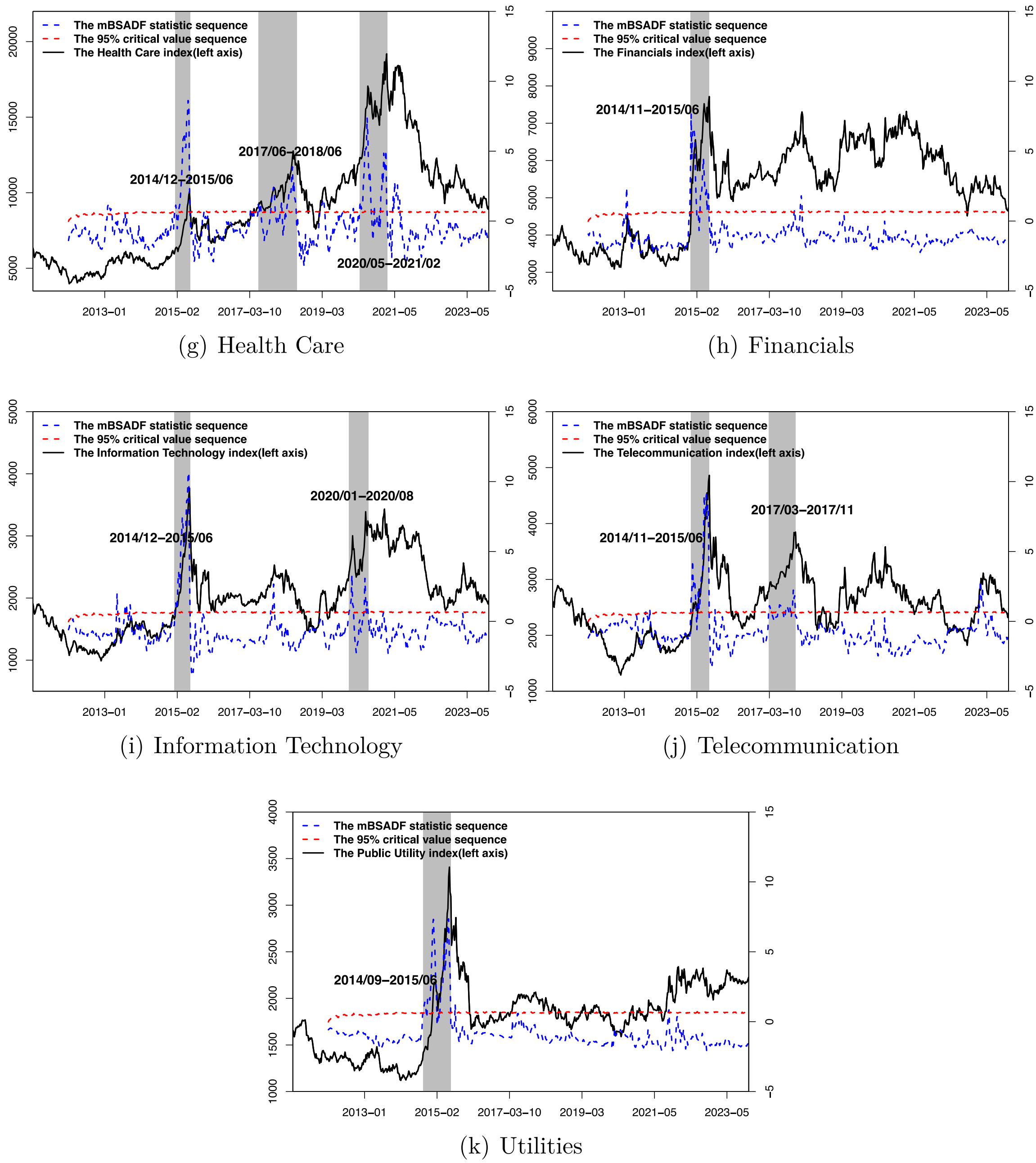


Fig. 1. Explosive bubble periods in the Chinese stock market and at the sector level.
Note: This figure illustrates the price trajectories of the Chinese stock market (proxied by the CSI 300 index) and 10 sector indices from January 7, 2011, to December 29, 2023, along with the bubble detection results. The black solid line represents the index price trajectories, while the blue and red dashed lines correspond to the mBSADF statistic and the corresponding finite sample critical value series. The gray shaded areas represent bubble periods for each index, identified by the mBSADF statistic exceeding and subsequently falling below the critical value (Eq. (4)). Only bubble periods with a duration of at least 3 months are reported, to exclude short-lived market fluctuations.

captures the 2017–18 episode, but fails to identify the 2020–21 bubble. Notably, the CSI 300 index rose from 3971.34 in June 2020 to 5417.65 in February 2021, a 35.4% increase within eight months. The results for sectoral indices are broadly consistent with those of the CSI 300 (see Table B.1), with the exception of Consumer Staples, where the BSADF provides limited evidence of the latter two bubbles.

We further examine whether structural characteristics shape sectoral bubble susceptibility. As shown in Table B.2, sectors with a higher share of state-owned enterprises (SOEs) (e.g., Utilities, Energy, Financials) appear more resilient to bubbles, while those with fewer SOEs (e.g., Health Care, Consumer Staples) are more prone to speculative surges. This pattern may reflect the distinctive nature of SOEs, whose behavior is often guided by policy objectives rather than short-term profit maximization. Their more stable financial performance may reduce the likelihood of investor speculation. By contrast, sectors dominated by private firms may attract greater speculative trading during periods of policy stimulus or technological optimism. This evidence is consistent with the economic reasoning behind Hypothesis 1: differences in sector-level fundamentals, ownership structure, and regulatory exposure help explain why bubble dynamics vary across sectors.



(g) Health Care (h) Financials

(i) Information Technology (j) Telecommunication

(k) Utilities

Fig. 1. (*continued*).

Taken together, these results support Hypothesis 1 by showing that bubbles in China's stock market are not uniformly synchronized across sectors. Instead, they tend to be structural, emerging in specific sectors under favorable policy, sentiment, or liquidity conditions. This motivates the next step of our analysis: to identify the key drivers of sector-level bubble occurrence, and to assess their relative explanatory relevance.

4.2. Influencing factors of bubble occurrence

4.2.1. Estimation results of the clog–log model

This section examines Hypothesis 2 by examining whether sectoral bubble occurrence is associated with investor behavior, sectoral fundamentals, and macroeconomic conditions. Specifically, we use the clog–log model in Eq. (5) to estimate the marginal effects of each category of determinants and assess whether their effects differ across sectors.[10] The dependent variable is a binary indicator, equal to 1 during bubble periods and 0 otherwise. Column (1) in Table 5 reports results for the aggregate market (CSI 300 index), while columns (2)–(11) show sector-level estimates, revealing heterogeneity in sector responses to potential bubble drivers. Table 5 displays the average marginal effects from the clog–log model, as shown in Eq. (6), which quantify how changes in these factors affect the probability of bubble occurrence.

The results yield several implications for Hypothesis 2. First, investor-behavior variables provide the clearest evidence consistent with this hypothesis. As shown in Table 5 (1), investor behavior variables are

[10] Descriptive statistics for the underlying factors during the three bubble and the non-bubble periods are also reported; see Appendix and Table B.3 for details.

Table 3
Descriptive statistics of potential influencing factors for bubble emergence.

	Mean	Median	Min	Max	Std.Dev.	IPS	LLC	N
ATR	1.026	0.946	0.192	4.380	0.413	-12.90^{***}	-3.22^{***}	6380
QS	−0.118	−0.118	−0.304	−0.026	0.048	-2.46^{***}	-1.42^{*}	6380
$Sent$	0.010	0.000	−1.360	1.730	0.513	-13.01^{***}	-6.82^{***}	6380
ROE	0.087	0.083	−0.049	0.309	0.054	-12.76^{***}	-3.12^{***}	6380
$Insti$	70.10	68.74	38.30	96.13	11.54	-11.27^{***}	-5.24^{***}	6380
$Asset$	0.036	0.031	−0.394	0.386	0.035	-14.01^{***}	-9.88^{***}	6380
EPU	144.15	141.10	98.36	234.52	28.49	-14.82^{***}	-9.16^{***}	6380
$rGDP$	1.570	1.700	−10.40	11.40	2.284	-22.30^{***}	-24.03^{***}	6380
$dShibor$	−0.004	0.001	−2.461	1.520	0.233	-44.29^{***}	-21.17^{***}	6380

Note: This table presents descriptive statistics for all potential determinants of bubbles, based on 580 weekly observations from January 6, 2012, to December 29, 2023, covering the CSI 300 and 10 sector indices (6380 observations per variable). Investor behavior factors (ATR, QS, $Sent$) correspond to abnormal turnover rate, quoted spread, and investor sentiment. Sectoral fundamentals (ROE, $Insti$, $Asset$) measure return on equity, institutional ownership, and asset growth rate. Macroeconomic factors (EPU, $rGDP$, $Shibor$) represent economic policy uncertainty, GDP growth, and the 1-week Shanghai Interbank Offered Rate. Panel unit root test results (IPS and LLC) are reported under the null of a unit root. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 4
Detected bubble duration of CSI 300 and sector indices.

	Period 1	Period 2	Period 3
CSI 300	2014/11-2015/06 8M	2017/09-2018/02 6M	2020/06-2021/02 9M
Energy	2014/11-2015/06 8M	/	/
Materials	2014/11-2015/06 8M	2017/07-2017/09 3M	2020/07-2021/09 15M
Industrials	2014/10-2015/06 9M	/	2020/11-2021/02 4M
Consumer Discretionary	2014/11-2015/06 8M	2017/06-2018/02 9M	2020/07-2021/02 8M
Consumer Staples	2015/03-2015/06 4M	2017/06-2018/03 10M	2019/03-2021/07 29M
Health Care	2014/12-2015/06 7M	2017/06-2018/06 13M	2020/05-2021/02 10M
Financials	2014/11-2015/06 8M	/	/
Information Technology	2014/12-2015/06 7M	/	2020/01-2020/09 9M
Telecommunication	2014/11-2015/07 9M	2017/03-2017/11 9M	/
Utilities	2014/09-2015/06 10M	/	/

Note: This table summarizes the bubble detection results based on mBSADF. The results are presented as the start and end dates and the durations of each explosive event for the CSI 300 and sector indices. All reported bubble periods last no less than 3 months to exclude short-lived market fluctuations. For comparison purposes, the explosive events within the sample period are categorized into three main phases.

positively associated with the probability of bubble occurrence in the aggregate market. Higher speculative trading, greater liquidity, and stronger investor optimism all increase the likelihood of bubbles, while the interaction between ATR and QS suggests that liquidity can further amplify speculative trading. By contrast, sectoral fundamentals and macroeconomic variables display much weaker effects, with GDP growth being the only macroeconomic variable that shows a clear positive association. In terms of economic magnitude, speculation (ATR) is the most influential factor: a one-standard-deviation increase raises the probability of bubble occurrence by 13.8 percentage points, followed by sentiment ($Sent^{\perp}$) and liquidity (QS). These results indicate that aggregate market bubbles in China are more closely related to investor behavior than to macroeconomic or sectoral fundamentals.

Second, the role of sectoral fundamentals is more sector-specific. As shown in columns (2)-(11) of Table 5, investor behavior generally retains a significant positive association with bubble occurrences across most sectors. By contrast, sectoral fundamentals show mixed and sector-specific effects. For example, ROE is positively related to bubbles in Consumer Discretionary but negatively related to bubbles in Information Technology, while $Asset$ does not exhibit a uniform pattern across sectors. Institutional ownership ($Insti$) also matters in selected sectors, where it tends to reduce bubble probability, suggesting that institutional participation may help dampen speculative pressures in some industries.

Third, macroeconomic conditions show relatively limited explanatory relevance. GDP growth shows a positive association with aggregate market bubbles, which is consistent with the cash-flow and growth-expectation channel. Macroeconomic factors, however, generally have limited explanatory power, indicating that sectoral bubbles in China are less consistently driven by aggregate economic conditions than by investor behavior and sector fundamentals.

The economic magnitudes further show that investor behavior affects sectors unevenly. Speculation and liquidity remain important drivers, but their effects are noticeably weaker in Financials. For example, a one-standard-deviation increase in ATR raises bubble probability by 15.9, 13.7, and 13.7 percentage points in Materials, Consumer Staples, and Consumer Discretionary, respectively, compared with 6.4 percentage points in Financials. A similar pattern is observed for liquidity, suggesting that speculative trading and liquidity pressures are weaker in more regulated sectors such as Financials. At the sectoral level, $Insti$ also emerges as a non-negligible factor, although its role is concentrated in a few sectors.

Our findings yield several key insights. First, investor behavior plays a dominant role in fueling bubble occurrences in the Chinese stock market, consistent with previous research (Scheinkman and Xiong, 2003; Carpenter et al., 2021). During periods of heightened volatility and optimism, retail investors may engage in herd behavior and short-term speculative trading (Allen et al., 2024). This collective movement increases trading pressure and may push stock prices away from fundamentals. Moreover, the interaction between speculation and liquidity further intensifies irrational trading, facilitating bubble formation (Wang and Wang, 2024). Second, sectoral fundamentals and macroeconomic conditions show weaker and more sector-specific associations with bubble dynamics, suggesting that bubble occurrences are not primarily driven by firm performance, earnings expectations, or broader economic conditions. This challenges the traditional view that earnings expectations and economic growth forecasts are primary drivers of asset bubbles (Sornette, 2009; Peiro, 2016), particularly in the context of China's market. Finally, we document substantial sectoral heterogeneity. Sectors characterized by high growth potential and liquidity, such as Consumer Staples and Materials, are more prone to investor behavior-driven booms, while Financials and Energy, often subject to tighter regulation and profit stability, show greater resistance to price deviations and lower bubble risks.

These patterns are consistent with Hypothesis 2: bubble occurrence in China's stock market is more strongly associated with investor behavior than with corporate fundamentals or macroeconomic

Table 5
Clog–log average marginal effects – CSI 300 and sector indices.

	CSI300 (1)	EN (2)	RM (3)	IND (4)	CD (5)	CS (6)	HC (7)	FIN (8)	IT (9)	TC (10)	PU (11)
ATR	0.138***	0.067***	0.159***	0.110***	0.137***	0.137***	0.076***	0.065***	0.052***	0.096***	0.139***
	(0.017)	(0.018)	(0.016)	(0.016)	(0.016)	(0.017)	(0.018)	(0.020)	(0.015)	(0.017)	(0.017)
QS	0.063***	0.034**	0.142***	0.044***	0.080***	0.191***	0.073***	0.029*	0.054***	0.049***	0.033*
	(0.019)	(0.017)	(0.021)	(0.017)	(0.020)	(0.019)	(0.016)	(0.017)	(0.017)	(0.018)	(0.018)
$Sent^{\perp}$	0.099***	0.021	0.049**	0.041*	0.116***	0.062***	0.048**	0.011	0.018	0.036**	−0.024
	(0.023)	(0.022)	(0.020)	(0.022)	(0.020)	(0.021)	(0.022)	(0.022)	(0.020)	(0.018)	(0.021)
$ATR * QS$	0.060***	0.047**	0.144***	0.047**	0.097***	0.132***	0.054***	−0.043**	0.038*	−0.016	−0.064***
	(0.023)	(0.020)	(0.021)	(0.024)	(0.018)	(0.020)	(0.018)	(0.019)	(0.020)	(0.018)	(0.023)
$ATR * Sent^{\perp}$	−0.043	0.037*	0.034	−0.009	0.091***	−0.020	0.016	−0.036	0.096***	0.130***	0.064***
	(0.027)	(0.021)	(0.022)	(0.023)	(0.022)	(0.020)	(0.019)	(0.027)	(0.016)	(0.020)	(0.018)
ROE	0.010	−0.029*	−0.015	−0.001	0.034**	−0.011	−0.026	0.017	−0.038**	−0.007	0.015
	(0.018)	(0.018)	(0.020)	(0.017)	(0.018)	(0.020)	(0.017)	(0.018)	(0.019)	(0.016)	(0.016)
$Insti$	0.004	−0.014	0.013	−0.017	−0.049**	−0.038*	−0.076***	−0.005	0.004	−0.001	−0.044***
	(0.020)	(0.018)	(0.017)	(0.016)	(0.021)	(0.021)	(0.015)	(0.017)	(0.017)	(0.018)	(0.017)
$Asset$	0.031*	−0.032*	0.016	−0.041**	0.031*	0.028*	−0.028*	0.089***	−0.003	−0.024	−0.005
	(0.018)	(0.017)	(0.019)	(0.019)	(0.018)	(0.017)	(0.016)	(0.018)	(0.020)	(0.016)	(0.017)
EPU	0.000	0.002	−0.022	0.002	0.004	−0.053***	−0.020	0.008	0.008	−0.027*	−0.027*
	(0.017)	(0.016)	(0.017)	(0.017)	(0.018)	(0.016)	(0.017)	(0.017)	(0.017)	(0.016)	(0.016)
$rGDP$	0.028**	−0.001	0.010	0.011	0.000	0.027	0.066***	0.013	0.004	0.007	0.002
	(0.014)	(0.015)	(0.015)	(0.015)	(0.016)	(0.019)	(0.015)	(0.014)	(0.022)	(0.015)	(0.018)
$dShibor$	0.053	0.036	0.080	0.066	0.098	0.069	0.060	0.037	−0.027	0.041	0.041
	(0.069)	(0.066)	(0.070)	(0.070)	(0.071)	(0.078)	(0.067)	(0.067)	(0.063)	(0.076)	(0.065)
Obs	580	580	580	580	580	580	580	580	580	580	580

Note: The table reports the average marginal effects as in Eq. (6) for each potential driver. Columns (1)–(11) correspond to CSI 300, Energy (EN), Materials (RM), Industrials (IND), Consumer Discretionary (CD), Consumer Staples (CS), Health Care (HC), Financials (FIN), Information Technology (IT), Telecommunication (TC), and Utilities (PU). The dependent variable equals 1 if the index is in a bubble period and 0 otherwise. Potential drivers include investor behaviors (ATR: abnormal turnover rate, QS: quoted spread, $Sent$: sentiment), sectoral fundamentals (ROE: return on equity, $Insti$: institutional ownership ratio, $Asset$: asset growth rate), and macroeconomic factors (EPU, $rGDP$: GDP Growth Rate, $dShibor$: 1-week Shanghai Interbank Offered Rate). All variables are standardized within sectors by subtracting their mean and dividing by the standard deviation, $x_{i,t}^{std} = (x_{i,t} - \bar{x}_{i,t})/(s_{x_{i,t}})$, to facilitate comparison. Standard errors are in parentheses, and *, **, *** denote significance at 10%, 5%, and 1%, respectively.

conditions. This pattern reflects the retail-dominated and speculation-sensitive structure of China's market, where institutional constraints may be less effective in curtailing speculative trading accompanied by excessive liquidity. Additionally, pronounced sectoral heterogeneity reveals the structural preferences of speculative capital, illustrating how it selectively targets industries perceived as sensitive to policy shifts or technological narratives.

4.2.2. Examining the relative importance of influencing factors

To complement the clog–log analysis, we use SHAP (Shapley additive explanation) values (Lundberg and Lee, 2017) to assess the relative importance of factors driving sectoral bubble occurrence. Rather than focusing only on marginal effects, SHAP ranks investor behavior, sectoral fundamentals, and macroeconomic conditions according to their average contribution to predicted bubble probabilities. This prediction-based perspective, which has been widely used in financial applications (Erel et al., 2021; Goodell et al., 2023; Aleti et al., 2025), provides a concise way to evaluate whether investor behavior remains the most important group of drivers.

Fig. 2 reports the mean absolute SHAP values (mean |SHAP|) of the candidate factors across the indices, reflecting their average contribution to the probability of bubble occurrence. For the overall market (CSI 300 index), investor behavior clearly dominates bubble dynamics: speculation (ATR), liquidity (QS), and sentiment ($Sent$) contribute 9.8, 8.8, and 8.8 percentage points, respectively, whereas fundamentals and macroeconomic factors contribute only 0.8–2.6 percentage points. This sharp contrast is consistent with Hypothesis 2, which expects investor behavior to exhibit greater explanatory relevance than sectoral fundamentals or macroeconomic conditions.

Regarding sector-level rankings, notable heterogeneity emerges. Speculation (ATR) and liquidity (QS) consistently rank among the most influential drivers across most sectors, confirming that investor behavior remains central to sectoral bubble formation. Sentiment ($Sent$) is also important in several sectors, especially those more exposed to investor attention and growth narratives. By contrast, sectoral fundamentals and macroeconomic conditions generally play a secondary role, although they become more relevant in selected sectors. This pattern is consistent with Hypothesis 2: fundamentals and macroeconomic conditions matter in some sectors, but investor behavior remains more important in predicting bubble occurrence.

Table 6 further confirms this pattern. ATR ranks among the top two factors for the overall market and most sectors, while macroeconomic variables (EPU, $rGDP$, and $dShibor$) generally appear near the bottom of the rankings. The main exceptions are Health Care and Financials, where the most influential factors are related to sectoral fundamentals rather than investor behavior. Overall, the SHAP results reinforce the clog–log findings: investor behavior is the leading driver of bubble occurrence, but sectoral fundamentals can matter in industries where ownership structure, asset expansion, or regulatory characteristics affect exposure to speculative trading.

Overall, the SHAP value analysis further supports Hypothesis 2 by showing that investor behavior has the greatest explanatory relevance for bubble prediction, while sectoral fundamentals and macroeconomic conditions play secondary but non-negligible roles. This reinforces the view that China's sectoral bubbles are more closely related to speculative trading, liquidity, and sentiment than to fundamentals or macroeconomic conditions alone.

4.3. Bubble contagion effects

This section examines Hypothesis 3 by assessing whether sectoral bubbles transmit across industries and to the aggregate market through time-varying contagion channels. Specifically, we employ the TVGC test to identify lead–lag causal relationships between sectors and the aggregate market, as well as among sectors. By detecting significant causal linkages that overlap with the previously identified bubble periods, we identify potential contagion effects among bubbles. The minimum window is set as $\lceil Tf_0 \rceil = \lceil 0.2\sqrt{T} \rceil = 51$ observations, as in Eq. (9). The test is conducted under the assumption of homoskedastic residuals within a $VAR(1)$ framework. Robust critical values are assessed via a 500-replication bootstrap procedure following Shi et al. (2018).

Fig. 3 reports the results of the time-varying Granger causality tests, treating each sector in turn as a potential source of influence. The

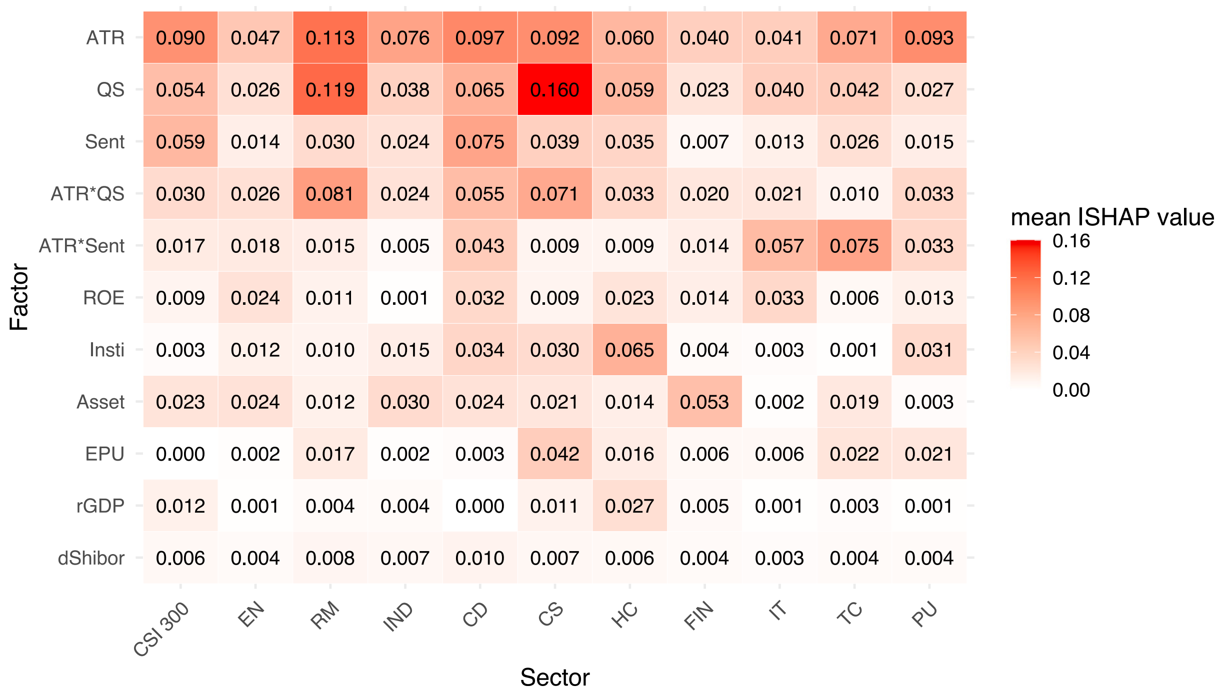


Fig. 2. SHAP importance heatmap of influencing factors.

Table 6
Ranking the importance of influencing factors.

Order	CSI 300	EN	RM	IND	CD	CS	HC	FIN	IT	TC	PU
1	ATR	ATR	QS	ATR	ATR	QS	Insti	Asset	ATRS	ATRS	ATR
2	Sent	QS	ATR	QS	Sent	ATR	ATR	ATR	ATR	ATR	ATRQ
3	QS	ATRQ	ATRQ	Asset	QS	ATRQ	QS	QS	QS	QS	ATRS
4	ATRQ	ROE	Sent	Sent	ATRQ	EPU	Sent	ATRQ	ROE	Sent	Insti
5	Asset	ATRS	EPU	ATRQ	ATRS	Sent	ATRQ	ATRS	ATRQ	EPU	QS
6	ATRS	Sent	ATRS	Insti	Insti	Insti	rGDP	ROE	Sent	Asset	EPU
7	rGDP	Insti	Asset	dShibor	ROE	Asset	ROE	Sent	EPU	ATRQ	Sent
8	ROE	Asset	ROE	ATRS	Asset	rGDP	EPU	EPU	dShibor	ROE	ROE
9	dShibor	Insti	dShibor	rGDP	dShibor	ROE	Asset	rGDP	Insti	dShibor	dShibor
10	Insti	EPU	dShibor	EPU	EPU	ATRS	ATRS	Insti	Asset	rGDP	Asset
11	EPU	rGDP	rGDP	ROE	rGDP	dShibor	dShibor	dShibor	rGDP	Insti	rGDP

Note: This table reports the importance rankings of explanatory factors for each index, based on their mean absolute SHAP values. The orders from 1 to 11 indicate decreasing levels of importance. $ATRQ$ denotes $ATR * QS$, and $ATRS$ represents $ATR * Sent^{\perp}$.

brown shaded areas denote periods of significant causal relationships, while the gray shaded areas represent bubble periods identified in Section 4.1. For instance, Fig. 3(a) illustrates the influence of the Energy sector on the overall market (proxied by the CSI 300 index) and the remaining nine sectors. This design allows us to distinguish two forms of contagion corresponding to Hypothesis 3: sector-to-market transmission, where sectoral signals precede or coincide with aggregate market bubbles, and inter-sectoral transmission, where bubbles propagate across industries through economic, financial, or investor-based linkages.

4.3.1. Sector-to-market contagion analysis

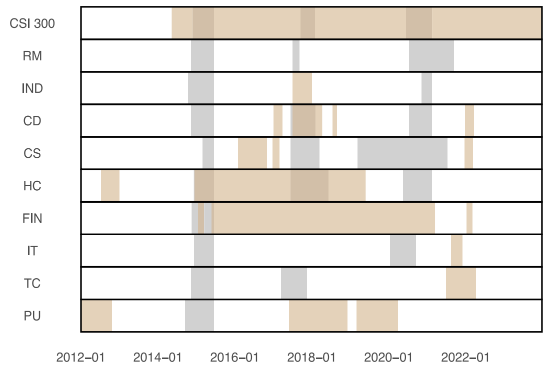
The top rows of Fig. 3(a)–(j) show the unidirectional influence of each sector on the overall market. Results show that sector-to-market causal linkages are closely aligned with market bubble episodes, although their onset and persistence differ markedly across sectors.[11] The test results highlight several key findings.

First, most sector-to-market causal relationships are concentrated in 2017–2018, whereas during the 2014–2015 bubble episode, only Energy (Fig. 3(a)), Materials (Fig. 3(b)) and Telecommunication (Fig. 3(i)) exhibit significant contagion effects. In conjunction with the bubble detection results in Table 4, it can be observed that the 2014 – 2015 sectoral bubbles largely coincide with, or lagged behind, the overall market bubble. This corroborates the view that the bubble during this period is primarily propelled by an environment characterized by widespread irrational sentiment and excessive liquidity, rather than sector-specific dynamics (Yang and Ferrer, 2023).

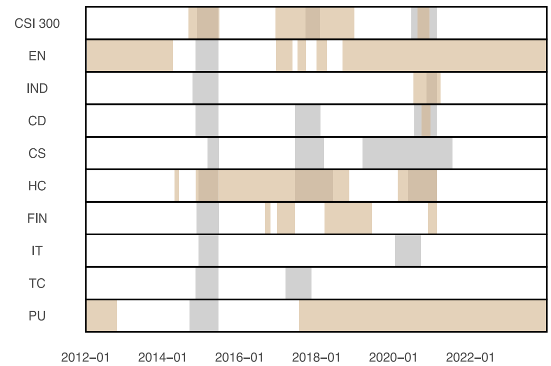
Second, Industrials and Utilities display weak or negligible transmission to the aggregate market, while Financials shows a more mixed pattern. Its direct sector-to-market effect is not always the strongest, but its importance becomes more evident in the cross-sector contagion results. This is consistent with the role of the financial sector in liquidity provision, leverage, and market-wide risk appetite. By contrast, Industrials and Utilities are more insulated due to the relative stability of their earnings, regulatory constraints, and lower sensitivity to speculative trading (Li and Huang, 2024). These results suggest that contagion may operate through both direct sector-to-market effects and indirect cross-sector linkages.

These findings are consistent with the early-warning implication of Hypothesis 3. Several sector-to-market contagion effects precede

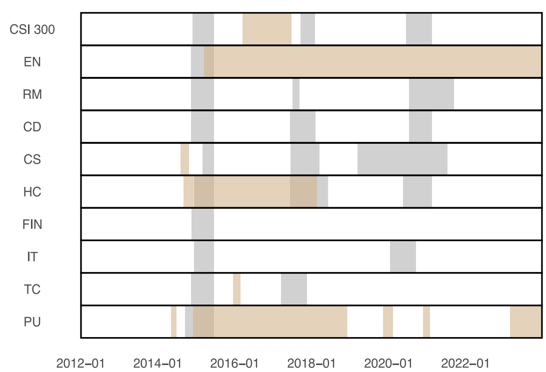
[11] In contrast, Fig. 3(k) shows the reverse direction, from the CSI 300 to sectors, revealing negligible bubble-aligned causal segments, except for the Energy sector. This asymmetry suggests that bubble dynamics often originate at the sectoral level before spreading to the broader market.



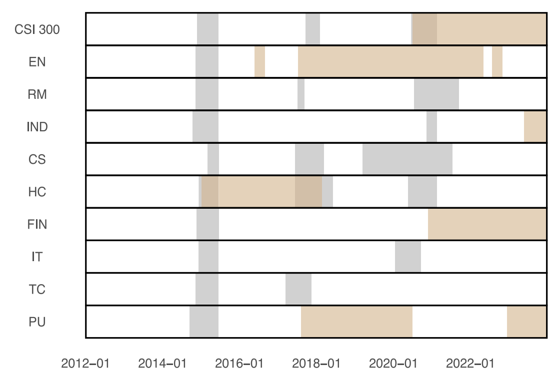
(a) Energy



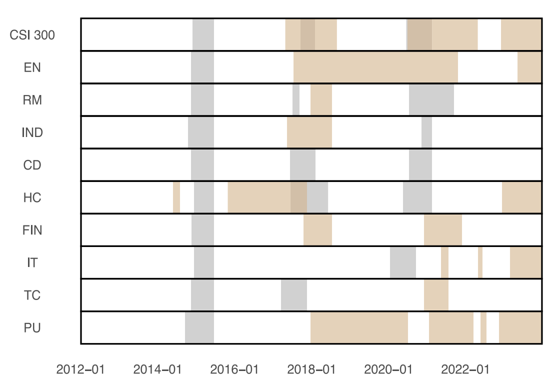
(b) Materials



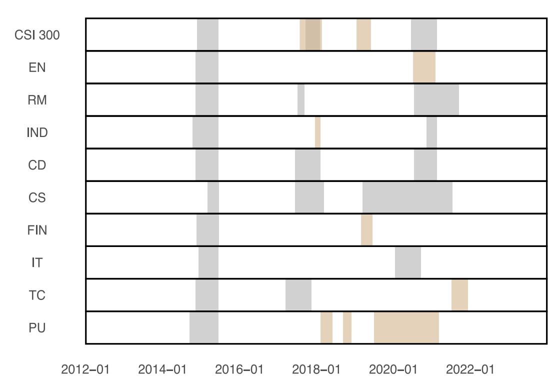
(c) Industrials



(d) Consumer Discretionary



(e) Consumer Staples



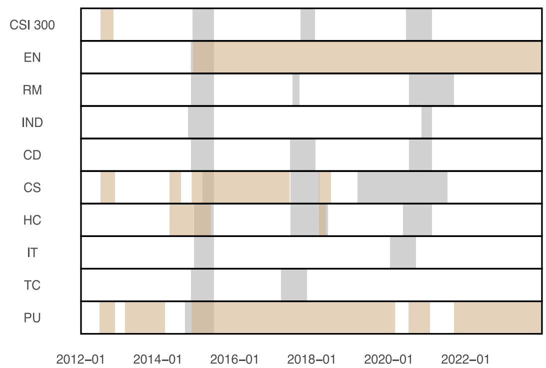
(f) Health Care

Fig. 3. The bubble contagion effect from sectors to the Chinese stock market.
Note: This figure presents the TVGC results among the CSI 300 and the 10 sector indices (a total of 11 indices) from January 7, 2011, to December 29, 2023. Each subplot displays the unidirectional causal effects from a specific index to the other 10 indices, with each row representing one affected index (row names indicate the affected index, and abbreviations follow those defined in the notes to Table 5). Brown shaded areas indicate periods of significant Granger causality, while gray shaded areas denote the bubble periods of the affected indices; the overlap of the two shaded regions reflects bubble contagion effects. Only causality episodes lasting more than two months are reported to exclude short-term fluctuations.

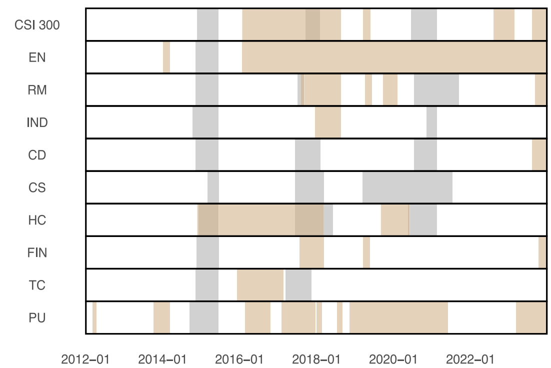
the emergence of overall market bubbles. For instance, the contagion effects of the Consumer Staples begin as early as March 2017 (bottom row in Fig. 3(e)), well before the onset of sectoral and market bubbles in July and November 2017, respectively (Table 4). This pattern is economically intuitive: when speculative pressure first concentrates in a sector, investors may subsequently reallocate capital, extrapolate sectoral narratives, or revise market-wide risk perceptions, allowing local bubble dynamics to diffuse into the aggregate index. From a policy perspective, real-time identification of such signals would enable regulators to adopt targeted liquidity control and intervention, thereby curbing risk escalation and enhancing overall market stability.

4.3.2. *Inter-sectoral contagion analysis*

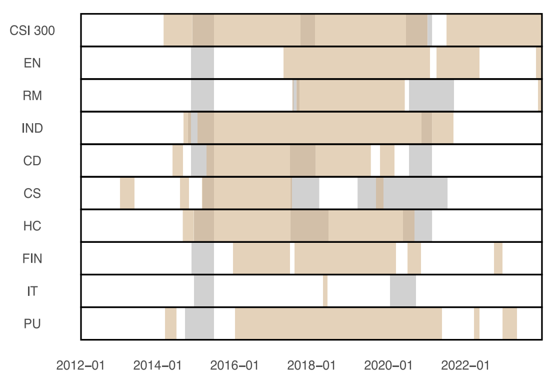
A granular understanding of inter-sectoral spillover dynamics is essential for constructing effective early warning systems for bubble formation. The top 9 rows of Fig. 3(a)–(j) visualize the time-varying unidirectional effects of each sector on the other 9, collectively forming



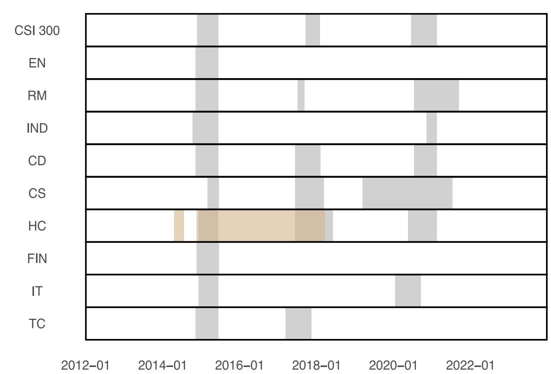
(g) Financials



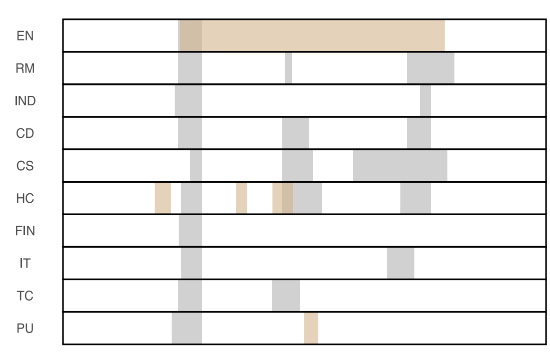
(h) Information Technology



(i) Telecommunication



(j) Utilities



(k) CSI 300

Fig. 3. (*continued*).

a dynamic inter-sectoral contagion network. Given the complexity of these results, several key insights emerge.

First, the Telecommunication sector consistently exerts significant unidirectional influence on a broad set of other sectors, while it remains largely insulated from external shocks. This asymmetry can be attributed to the sector's rising prominence amid China's digital economy boom, which has accelerated corporate digital transformation and strengthened the spillover effects of the Telecommunication sector, particularly in emerging industries such as cloud computing and big data analytics (J. Liu et al., 2023). In addition, stringent regulatory oversight of technology and information-related industries has likely dampened their vulnerability to external shocks (Hua and Huang, 2021). The Financials sector emerges as another major source of bubble contagion. Its centrality stems from high leverage, strong interconnectivity, and core role in liquidity provision, which collectively enhance its capacity to transmit bubbles (Tobias and Brunnermeier, 2016).

In contrast, the Health Care sector predominantly functions as a recipient rather than a source of contagion. This asymmetry is largely due to its defensive investment characteristics, lower exposure to speculative capital, and peripheral network position, which together limit its potential as a risk transmitter (Asgharian et al., 2022). Furthermore, the spillover from the Energy and Materials sectors to Consumer Staples and Industrial sectors, and subsequently to Utilities (Energy/Materials-Consumer Staples/Industrials-Utilities), reflects a typical supply-chain–style risk transmission pathway: shocks originating in upstream commodity sectors propagate through cost channels to midstream sectors and eventually diffuse to downstream industries. Such transmission pathways echo real-world input–output structures, where upstream price volatility can amplify sectoral risk along production networks.

The inter-sectoral results further support Hypothesis 3 by showing that bubble contagion is not limited to sector-to-market transmission but also operates through cross-industry linkages. Telecommunications and Financials emerge as important transmitters, whereas Health Care, Industrials, and Utilities display more limited contagion potential. This heterogeneity is consistent with economically meaningful transmission channels, rather than uniform market-wide co-movement alone.

In summary, pronounced sector-to-market and cross-sector contagion patterns exist during bubble episodes in the Chinese stock market. Some sector-to-market signals appear before the recipient sectors or the aggregate market enter bubble periods, suggesting their value as early warning indicators. At the same time, contagion dynamics are heterogeneous across sectors. Telecommunications serves as an important transmitter, consistent with the role of digital-economy narratives and investor attention in driving cross-sector spillovers. Financials also acts as a key contagion source, mainly through its links with liquidity provision, leverage, and market-wide risk appetite. In contrast, sectors subject to strict regulatory oversight or defensive investment characteristics, such as Industrials, Utilities, and Health Care, display more limited contagion potential. These findings support differentiated monitoring strategies that account for both sector-specific vulnerability and cross-sector transmission channels.

4.4. Discussion and implications

The empirical evidence provides a coherent picture of the hypotheses and economic mechanisms developed in Section 2. The bubble detection results support Hypothesis 1 by showing that bubble episodes differ substantially across sectors in timing, duration, and intensity. The clog–log and SHAP analyses provide evidence consistent with Hypothesis 2, indicating that investor behavior has the greatest explanatory relevance, while sectoral fundamentals and macroeconomic conditions play more sector-specific roles. The TVGC results further support Hypothesis 3 by showing that sectoral bubbles can transmit across industries and to the aggregate market, with some contagion signals preceding market-wide bubble episodes.

Building on this evidence, this study sheds light on sectoral bubble formation and contagion dynamics in the Chinese stock market by connecting bubble occurrence to investor behavior, sectoral characteristics, and cross-sector transmission. The results suggest that bubbles in China do not necessarily emerge as aggregate market phenomena from the outset. Rather, speculative pressures may first build up within particular sectors when investor attention, liquidity, and valuation optimism become concentrated in industries with salient growth prospects or policy-related narratives. This interpretation is consistent with China's market environment, where high retail participation, short-selling constraints, and limited arbitrage capacity make prices more sensitive to sentiment-driven trading and sector-specific expectations.

The dominant role of investor behavior is central to this interpretation. Bubble formation is more strongly related to trading behavior, liquidity, and sentiment than to sectoral fundamentals or macroeconomic variables alone. This does not imply that fundamentals are irrelevant; instead, they may shape the sectors in which speculative trading is more likely to concentrate, while investor behavior helps explain why price dynamics become explosive in the short run. The sectoral evidence also shows that speculative pressures are unevenly distributed across industries: high-growth sectors and sectors exposed to salient narratives are more likely to attract investor attention and capital inflows, whereas regulated or defensive sectors tend to display weaker bubble dynamics.

Sectoral heterogeneity further shapes how bubble-related risks are transmitted. Once sectoral bubbles arise, they may spread through capital reallocation, common investor behavior, valuation spillovers, and sectoral linkages. The roles of Financials and Telecommunications are economically intuitive: Financials is closely related to liquidity provision, leverage, credit conditions, and market-wide risk appetite, while Telecommunications is associated with technology-related expectations and digital-economy narratives that attract investor attention and speculative capital. By contrast, sectors such as Industrials and Utilities are more insulated because of relatively stable earnings, regulatory constraints, and defensive investment characteristics. These patterns help explain why sectoral contagion signals tend to precede market-wide bubbles and why sector-level monitoring can provide information beyond aggregate market indicators.

Overall, the combination of investor-driven sectoral bubbles and subsequent cross-sector transmission helps explain the emergence of market-wide bubble risks. These findings suggest the limitations of relying solely on aggregate market indicators for systemic risk monitoring. From a policy perspective, they underscore the importance of sector-specific monitoring and differentiated regulatory responses. Since sectoral contagion signals tend to precede market-wide bubbles, regulatory authorities may benefit from tracking sectoral-level bubble indicators such as abnormal turnover, liquidity conditions, and investor sentiment. Targeted measures aimed at curbing excessive speculation in key sectors could help contain localized bubbles before they contribute to broader market instability, thereby enhancing the resilience and stability of China's capital markets.

4.5. Robustness checks

To validate the robustness of our findings, we undertake a series of checks that include the implementation of an alternative bubble detection technique and the application of a different probabilistic model. The LPPLS test examines whether the sectoral heterogeneity documented in Hypothesis 1 is robust to an alternative bubble dating method, while the logit model assesses whether the conclusions regarding the determinants of bubble occurrence in Hypothesis 2 are sensitive to the choice of probabilistic specification.

4.5.1. Alternative bubble detection technique

Accurate estimation of bubble duration is crucial for the robustness of our conclusions. We employ the log-periodic power-law singularity (LPPLS) model as a robustness check for our bubble detection results. Based on the premise that financial bubbles exhibit super-exponential growth with log-periodic oscillations, the LPPLS model captures nonlinear feedback and critical point behavior, making it effective in identifying bubble emergence and collapse. LPPLS has been widely used to study bubble dynamics across asset classes (See, e.g., Assaf et al., 2024; Huang and Wang, 2024). We adopt the stable and robust version by Filimonov and Sornette (2013), which improves detection accuracy under volatile market conditions.

Table B.4 summarizes LPPLS bubble detection results, capable of identifying both expansion (positive bubbles) and recovery (negative bubbles) phases. For this study, we focus on positive bubbles, aligning with the mBSADF findings in Table 4. LPPLS provides strong support for the 2014 – 15 episode. LPPLS identifies a bubble in the CSI 300 from November 2014 to June 2015, and almost all sector indices exhibit bubble episodes over a similar window, with most terminating around June 2015. For the later episodes, the LPPLS results suggest a more sector-specific pattern. The 2017–2018 episode is detected only in Consumer Staples, indicating that this phase was much less pervasive than the 2014–2015 market-wide bubble. By contrast, the 2020–2021 episode is identified in several sectors, including Materials, Consumer Discretionary, Consumer Staples, and Health Care, with most of these bubbles ending around February 2021.

Taken together, the LPPLS results support our main conclusion that the 2014–2015 episode was broad-based, whereas the 2017–2018 and 2020–2021 episodes were more sector-specific. This reinforces Hypothesis 1 by showing that the heterogeneity in sectoral bubble timing and coverage is not driven solely by the mBSADF dating procedure.

4.5.2. Alternative probabilistic model

The main advantage of employing the clog–log model over traditional probabilistic models lies in its ability to characterize the distribution of the dependent variable. In contrast, logit and probit assume a linear change in the log-odds of the dependent variable, widely applied approaches in bubble literature (See, e.g., Enoksen et al., 2020; Haykir and Yagli, 2022; Chang, 2024). To ensure comparability, we also re-estimate the relationship between bubble occurrence and its drivers using a logit model, specified as:

$$\begin{aligned} & log(\frac{P(bub_t = 1)}{1 - P(bub_t = 1)}) \\ & = \beta_1 ATR_t + \beta_2 QS_t + \beta_3 Sent_t^{\perp} + \beta_4 ATR_t * QS_t + \beta_5 ATR_t * Sent_t^{\perp} + \\ & \quad \beta_6 ROE_t + \beta_7 Insti_t + \beta_8 Asset_t + \beta_9 EPU_t + \beta_{10} rGDP_t \\ & \quad + \beta_{11} dShibor_t + e_t, \end{aligned} \tag{11}$$

Eq. (11) details the time series model used; the variables are defined as in Section 3.2 and standardized for comparability. We still focus our reporting on the average marginal effects, which are documented in Table B.5.

The marginal effects from logit and clog–log models are largely consistent in both direction and magnitude, reinforcing the robustness of our approach. Predominantly, investor behavior, particularly speculative activity measured by abnormal turnover rate (ATR), remains the key driver of bubble occurrences, while sectoral fundamentals and macroeconomic variables have relatively minor effects. Notably, the logit model tends to yield higher marginal effects than the clog–log model. For example, a one-standard-deviation increase in ATR raises bubble probability by 10 to 24.7 percentage points in the logit model (Table B.5) versus 5.2 to 15.9 points in the clog–log results (Table 5), reflecting the logit model's tendency to overestimate effects in rare-event contexts. This validates the choice of the clog–log model for capturing the asymmetrical distribution of rare events like financial bubbles. These results indicate that our main conclusion regarding the dominant role of investor behavior in sectoral bubble formation is not driven by the choice of probabilistic model.

These results reinforce the evidence for Hypothesis 2 by showing that investor behavior remains the most stable and economically meaningful predictor of bubble occurrence, while sectoral fundamentals and macroeconomic conditions play more limited and sector-specific roles. Thus, our main conclusion is not driven by the choice of probabilistic model.

5. Conclusion

This study extends the literature by revealing the dynamic evolution of sector-level bubbles in the Chinese stock market from January 2011 to December 2023. Using the bubble detection method proposed by Wang et al. (2023), which builds on and modifies the approach of Phillips et al. (2015a), we identify both market-wide and sector-specific bubble episodes. Our results confirm the well-known 2014–2015 bubble episode and subsequent crash and further uncover two underexplored bubble periods in 2017−2018 and 2020−2021. While the 2015 bubble occurs in all sectors, the later bubbles are confined to specific sectors, including Materials, Consumer Discretionary, and Health Care, thereby being labeled as "structural bubbles". Applying a complementary log–log model, we show that investor behavior is the primary driver of bubbles, particularly in high-growth sectors, whereas stable and regulated sectors like Financials, Energy, and Utilities, exhibit lower susceptibility to speculative activity and sentiment. By contrast, sectoral fundamentals and macroeconomic conditions exert limited roles and are significant only in a few sectors. Overall, these findings underscore the dominant role of investor behavior in shaping bubble dynamics in the Chinese stock market.

We further apply the time-varying Granger causality test of Shi et al. (2018) to examine potential bubble contagion effect across indices. The results indicate that some sectors act as significant one-way transmitters of bubble risk to the broader market. In particular, Financials and Telecommunications emerge as important contagion sources, consistent with their respective roles in liquidity provision, leverage, market-wide risk appetite, and technology-related narratives. These findings suggest that sectoral bubbles do not always remain localized; instead, they may contain early information about broader market instability when bubble-related signals spread across industries or to the aggregate market.

Overall, the findings suggest that sectoral bubbles in China are closely associated with the joint roles of investor behavior and sectoral characteristics, while their systemic relevance depends on whether bubble-related signals are transmitted across industries. The prominent role of investor behavior indicates that speculative trading, liquidity, and sentiment are central to the formation of sectoral bubbles, especially in sectors exposed to growth prospects or salient market narratives. At the same time, sectoral characteristics help explain why speculative pressures are concentrated in some industries but not others. The contagion results further show that cross-sector transmission channels can transform localized imbalances into broader market risks, especially when bubbles arise in sectors closely connected to liquidity conditions, investor attention, or market-wide risk perceptions.

These findings carry important implications for systemic risk monitoring and regulatory design. The prominent role of investor behavior suggests that valuation-based indicators alone may be insufficient for identifying bubble risk in China. Sector-level surveillance can complement aggregate market monitoring because contagion signals may originate within specific sectors before spreading to the broader market. Moreover, the systemic relevance of a sector depends not only on the intensity of its own bubble, but also on its position in the cross-sector transmission network. Regulatory attention should therefore account for both sector-specific vulnerabilities and cross-sector linkages. Such a framework may help identify early-warning signals and support more targeted interventions aimed at mitigating systemic risk.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

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Appendix A. Comparison between BSADF and mBSADF

The primary bubble detection method adopted in this paper is the modified BSADF (mBSADF), which has already been described in detail in the main text. Since the mBSADF and its traditional counterpart BSADF are highly similar in terms of the construction of test statistics, the date-stamping rules, and their asymptotic properties, this appendix revisits both methods to ensure transparency and clarity, serving as a supplement to the main text.

A.1. Hypotheses

Both BSADF and mBSADF are based on the following data generating process:

$$y_t = dT^{-\eta} + \rho y_{t-1} + u_t, t = 1, 2, \ldots, T, \tag{A.1}$$

where $dT^{-\eta}$ is an asymptotically negligible drift and the null and alternative hypotheses are:

$$H_0 : \rho = 1 \text{ (random walk)}, \quad H_a : \rho > 1 \text{ (explosive behavior)}$$

Table B.1
Bubble duration of CSI 300 and sector indices detected by BSADF method.

	Period 1	Period 2	Period 3
CSI 300	2014/11-2015/06 8M	2017/10-2018/01 4M	/
Energy	2014/12-2015/06 7M	/	/
Materials	2014/11-2015/06 8M	/	/
Industrials	2014/10-2015/06 9M	/	/
Consumer Discretionary	2014/12-2015/06 7M	/	2020/10-2021/02 5M
Consumer Staples	2015/03-2015/06 4M	2017/06-2018/03 10M	2019/03-2021/07 29M
Health Care	2015/03-2015/06 4M	/	/
Financials	2014/11-2015/06 8M	/	/
Information Technology	2014/12-2015/06 7M	/	/
Telecommunication	2014/12-2015/06 7M	/	/
Public Utility	2014/09-2015/06 10M	/	/

Note: This table summarizes the bubble detection results based on the BSADF method. The results are presented as the start and end dates and the durations of each explosive event for the CSI 300 and sector indices. All reported bubble periods last no less than 3 months to exclude short-lived market fluctuations. For comparison purposes, the explosive events within the sample period are categorized into three main phases.

Table B.2
The ranking of the SOE proportion.

2015		2017		2021	
Public Utility	1.00	Public Utility	1.00	Public Utility	1.00
Industrials	0.81	Energy	0.83	Energy	1.00
Energy	0.80	Industrials	0.74	Financials	0.59
Materials	0.79	Materials	0.61	Industrials	0.48
Financials	0.66	Financials	0.57	Materials	0.46
Consumer Discretionary	0.53	Consumer Staple	0.45	Telecommunication	0.45
Telecommunication	0.50	Telecommunication	0.43	Consumer Staple	0.36
Consumer Staple	0.43	Consumer Discretionary	0.30	Information Technology	0.26
Health Care	0.26	Health Care	0.27	Consumer Discretionary	0.20
Information Technology	0.24	Information Technology	0.20	Health Care	0.18

Note: This table shows the proportion of state-owned enterprises (SOEs) in each sector for the years 2015, 2017, and 2021. The SOE proportion is calculated by dividing the number of SOEs in each sector by the total number of firms in that sector. The sectors are ranked from highest to lowest based on the proportion of SOEs.

Table B.3
Descriptive statistics of potential influencing factors for bubble emergence.

	Bubble period 1			Bubble period 2			Bubble period 3			Other period		
	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max
ATR	1.954	0.955	4.380	1.092	0.444	2.066	1.167	0.403	2.794	0.960	0.192	3.365
QS	0.108	0.058	0.205	0.101	0.038	0.183	0.100	0.030	0.203	0.121	0.026	0.304
$Sent$	1.065	0.410	1.730	0.252	−0.290	0.690	0.592	0.030	1.370	−0.099	−1.360	1.240
ROE	0.067	0.007	0.186	0.096	−0.037	0.251	0.105	0.003	0.265	0.087	−0.048	0.309
$Insti$	66.44	42.22	93.85	70.79	45.10	96.10	70.45	53.06	93.94	70.26	38.30	96.13
$Asset$	0.045	−0.004	0.347	0.033	−0.012	0.101	0.048	−0.024	0.153	0.034	−0.394	0.386
EPU	144.2	132.9	110.2	111.3	101.9	128.3	147.8	129.3	169.3	146.0	98.36	234.5
$rGDP$	1.78	1.60	2.00	1.66	1.60	1.80	2.84	0.50	11.40	1.47	−10.40	11.40
$dShibor$	0.043	−0.190	1.342	−0.003	−0.073	0.035	0.042	−0.162	0.377	−0.010	−2.461	1.520

Note: This table provides brief descriptive statistics of the potential determinants of bubbles categorized by periods. The division between Bubble Periods 1-3 and Other Periods is based on the CSI 300 index, which corresponds to the first (November 2014 to June 2015), second (September 2017 to February 2018), and third (June 2020 to February 2021) bubble periods, as well as the remaining periods. ATR, QS, and $Sent$ represent factors of investor behaviors, corresponding to abnormal turnover rate, quoted spread, and investor sentiment, respectively. ROE, $Insti$, and $Asset$ represent sectoral fundamental factors, measuring return on equity, institutional ownership ratio, and asset growth rate, respectively. EPU, $rGDP$, and $Shibor$ account for macroeconomic factors, representing economic policy uncertainty, GDP growth rate, and the 1-week Shanghai Interbank Offered Rate, respectively. The original series for ROE, $Insti$, $Asset$, and $rGDP$ are quarterly, while EPU is monthly. All variables are converted to weekly frequency to ensure consistency across data frequencies.

A.2. Test statistics

Both BSADF and mBSADF are constructed from right-tailed augmented Dickey–Fuller (ADF) regressions estimated recursively. For each observation y_t, $t := \lfloor Tf \rfloor$, an ADF regression is first performed on subsamples with fraction starting point f_1 and endpoint f:

$$\Delta y_t = \hat{\alpha}_{f_1,f} + \hat{\beta}_{f_1,f} y_{t-1} + \sum_{j=1}^{p} \hat{\theta}^j_{f_1,f} \Delta y_{t-j} + u_t, t = \lfloor Tf_1 \rfloor, \dots, \lfloor Tf \rfloor, \tag{A.2}$$

Denote the corresponding ADF statistic as $ADF_{f_1,f} = \hat{\beta}_{f_1,f}/se(\hat{\beta}_{f_1,f})$, and rolls the starting point f_1 from $f-f_0$ to 0 to obtain $\{ADF_{f_1,f}\}_{f_1\in[0,f-f_0]}$, BSADF statistic at f is defined as

$$BSADF_f(f_0) = \sup_{f_1\in[0,f-f_0]} ADF_{f_1,f}, \tag{A.3}$$

where f_0 is the minimum sample window length. The mBSADF augments BSADF by adding a modification term based on the first difference Δy_t:

$$mBSADF_f(f_0) = \sup_{f_1\in[0,f-f_0]} ADF_{f_1,f} + \frac{d}{K} \sum_{t=\lfloor Tf \rfloor - K+1}^{\lfloor Tf \rfloor} m_t, f \in [f_0, 1], \tag{A.4}$$

where $m_t = \Delta y_t/\sigma_{\Delta y}$, $c > 0$ and a K is a moving window. This refinement improves responsiveness to short-lived explosiveness and reduces spurious signals. A series of statistics $\{BSADF_f(f_0)\}_{f\in[f_0,1]}$ and $\{mBSADF_f(f_0)\}_{f\in[f_0,1]}$ are obtained by rolling the endpoint f.

A.3. Date-stamping rules

Both BSADF and mBSADF adopt the same dating procedure. Given the sequence of test statistics, the estimated bubble origination and

Table B.4
Robustness check: Bubble duration of CSI 300 and sector indices detected by LPPLS method.

	Period 1	Period 2	Period 3
CSI 300	2014/11-2015/06 8M	/	/
Energy	2015/03-2015/06 4M	/	/
Materials	2014/11-2015/06 8M	/	2020/10-2021/02 5M
Industrials	2014/10-2015/06 9M	/	/
Consumer Discretionary	2014/12-2015/06 7M	/	2020/09-2021/02 6M
Consumer Staples	2015/03-2015/06 4M	2017/09-2018/02 6M	2020/06-2021/02 9M
Health Care	2015/03-2015/06 4M	/	2020/06-2020/10 5M
Financials	2014/11-2015/06 8M	/	/
Information Technology	2015/01-2015/06 6M	/	/
Telecommunication	2014/11-2015/07 9M	/	/
Public Utility	2014/11-2015/06 8M	/	/

Note: This table summarizes the bubble detection results based on the LPPLS method of Filimonov and Sornette (2013). The results are presented as the start and end dates and the durations of each explosive event for the CSI 300 and sector indices. All reported bubble periods last no less than 3 months to exclude short-lived market fluctuations. For comparison purposes, the explosive events within the sample period are categorized into three main phases.

Table B.5
Robustness check: Logit average marginal effects – CSI 300 and sector indices.

	CSI 300 (1)	EN (2)	RM (3)	IND (4)	CD (5)	CS (6)	HC (7)	FIN (8)	IT (9)	TC (10)	PU (11)
ATR	0.212***	0.107***	0.247***	0.167***	0.202***	0.199***	0.123***	0.099***	0.109***	0.161***	0.209***
	(0.025)	(0.025)	(0.020)	(0.022)	(0.020)	(0.021)	(0.023)	(0.030)	(0.021)	(0.022)	(0.023)
QS	0.097***	0.051**	0.214***	0.064***	0.1000***	0.239***	0.099***	0.037	0.076***	0.078***	0.035
	(0.023)	(0.022)	(0.025)	(0.022)	(0.023)	(0.020)	(0.019)	(0.024)	(0.022)	(0.022)	(0.023)
$Sent^{\perp}$	0.111***	0.032	0.067***	0.054*	0.166***	0.067***	0.073***	−0.011	0.049*	0.101***	−0.015
	(0.028)	(0.028)	(0.024)	(0.028)	(0.024)	(0.024)	(0.025)	(0.029)	(0.026)	(0.025)	(0.027)
$ATR * QS$	0.116***	0.083***	0.276***	0.074**	0.135***	0.180***	0.082***	−0.091***	0.057**	0.020	−0.098***
	(0.032)	(0.028)	(0.032)	(0.032)	(0.023)	(0.024)	(0.022)	(0.034)	(0.026)	(0.025)	(0.033)
$ATR * Sent^{\perp}$	−0.021	0.056*	0.041	0.012	0.124***	−0.029	0.014	−0.048	0.169***	0.191***	0.089***
	(0.038)	(0.029)	(0.034)	(0.034)	(0.030)	(0.024)	(0.024)	(0.038)	(0.026)	(0.027)	(0.024)
ROE	0.007	−0.042*	0.010	0.001	0.039*	−0.012	−0.027	0.039	−0.042*	−0.015	0.014
	(0.022)	(0.023)	(0.024)	(0.021)	(0.022)	(0.023)	(0.021)	(0.024)	(0.024)	(0.021)	(0.021)
$Insti$	0.014	−0.020	0.003	−0.017	−0.052**	−0.038	−0.107***	−0.003	0.003	−0.016	−0.074***
	(0.026)	(0.024)	(0.020)	(0.021)	(0.025)	(0.025)	(0.019)	(0.023)	(0.021)	(0.022)	(0.024)
$Asset$	0.045**	−0.042*	0.024	−0.053**	0.038*	0.037*	−0.047**	0.188***	−0.009	−0.021	−0.010
	(0.022)	(0.022)	(0.022)	(0.024)	(0.022)	(0.020)	(0.021)	(0.036)	(0.025)	(0.020)	(0.021)
EPU	−0.009	0.003	−0.032	0.006	0.006	−0.065***	−0.026	0.002	0.023	−0.027	−0.032
	(0.021)	(0.021)	(0.020)	(0.023)	(0.021)	(0.018)	(0.020)	(0.022)	(0.022)	(0.020)	(0.021)
$rGDP$	0.041**	−0.002	0.012	0.014	0.004	0.033**	0.096***	0.025	0.001	0.015	0.005
	(0.019)	(0.020)	(0.017)	(0.020)	(0.019)	(0.017)	(0.024)	(0.019)	(0.019)	(0.019)	(0.021)
$Shibor$	0.051	0.044	0.07	0.078	0.082	0.077	0.082	0.069	0.017	0.040	0.041
	(0.083)	(0.087)	(0.079)	(0.088)	(0.086)	(0.088)	(0.082)	(0.085)	(0.088)	(0.095)	(0.083)
Obs	580	580	580	580	580	580	580	580	580	580	580

Note: The table reports the average marginal effects for each potential driver. Columns (1)–(11) correspond to CSI 300, Energy (EN), Materials (RM), Industrials (IND), Consumer Discretionary (CD), Consumer Staples (CS), Health Care (HC), Financials (FIN), Information Technology (IT), Telecommunication (TC), and Utilities (PU). The dependent variable equals 1 if the index is in a bubble period and 0 otherwise. Potential drivers include investor behaviors (ATR: abnormal turnover rate, QS: quoted spread, $Sent$: sentiment), sectoral fundamentals (ROE: return on equity, $Insti$: institutional ownership ratio, $Asset$: asset growth rate), and macroeconomic factors (EPU, $rGDP$: GDP Growth Rate, $dShibor$: 1-week Shanghai Interbank Offered Rate). All variables are standardized within sectors by subtracting their mean and dividing by the standard deviation, $x_{i,t}^{std} = (x_{i,t} - \bar{x}_{i,t})/(s_{x_{i,t}})$, to facilitate comparison. Standard errors are in parentheses, and *, **, *** denote significance at 10%, 5%, and 1%, respectively.

termination dates $(\hat{f}_o, \hat{f}_c)$ are defined as

$$\hat{f}_o = \inf_{f\in[f_0,1]} \{ f : BSADF_f(f_0) > scv_f \},$$
$$\hat{f}_c = \inf_{f\in[\hat{f}_o+L_b,1]} \{ f : BSADF_f(f_0) < scv_f \}, \tag{A.5}$$

$$\hat{f}_o = \inf_{f\in[f_0,1]} \{ f : mBSADF_f(f_0) > mscv_f \},$$
$$\hat{f}_c = \inf_{f\in[\hat{f}_o+L_b,1]} \{ f : mBSADF_f(f_0) < mscv_f \}, \tag{A.6}$$

where scv_f and $mscv_f$ denote the 95% critical values of $BSADF_f(f_0)$ and $mBSADF_f(f_0)$, respectively. The constant L_b is a minimum duration requirement to avoid misclassifying short-lived noise as bubbles.

A.4. Asymptotic properties

According to Wang et al. (2023), the BSADF and mBSADF share the same asymptotic distribution under the null hypothesis H_0:

$$F_f(W, f_0) := \sup_{f_1\in[0,f-f_0]} \frac{\int_{f_1}^{f} \underline{W}(r)\, dW(r)}{\sqrt{\int_{f_1}^{f} \underline{W}(r)^2\, dr}}, \tag{A.7}$$

where $\underline{W}(r) = W(r) - \frac{1}{f-f_1}\int_{f_1}^{f} W(r)\, dr$ denotes the demeaned standard Brownian motion $W(r)$. Consequently,

$$BSADF_f(f_0) \Rightarrow F_f(W, f_0), \quad mBSADF_f(f_0) \Rightarrow F_f(W, f_0).$$

Under the alternative hypothesis H_a, both BSADF and mBSADF statistics diverge with positive probability, and the associated date-stamping algorithms deliver consistent estimators of the bubble origination and termination dates (f_o, f_c). The critical value requirements for consistency are identical across the two procedures.

Appendix B. Supplementary results and robustness checks

B.1. Bubble detection results using BSADF

See Table B.1.

B.2. Structural characteristics and sectoral bubble susceptibility

See Table B.2.

B.3. Description statistics of factors during bubble and non-bubble periods

To provide more intuition about bubble dynamics, we firstly analyze the performance of the three categories of influencing factors across bubble (November 2014 to June 2015, September 2017 to February 2018, June 2020 to February 2021) and non-bubble periods. Table A.3 shows that the average values of ATR during the three bubble periods (1.954, 1.092 and 1.167) are significantly higher than those during the remaining periods (0.960), particularly during bubble periods 1 and 3. Similarly, the average values of sentiment are also notably higher during the bubble periods (1.065, 0.252, 0.592), with remaining periods exhibiting the level of −0.099. The QS levels during the bubble periods are slightly higher compared to the remaining periods (−0.108, −0.101, −0.100 vs. −0.121). The sector fundamentals and macroeconomic conditions remain relatively stable across all periods, with the exception of the average $Asset$ and $dShibor$ during bubble periods 1 and 3. With this initial inspection, we conjecture that investor behaviors, rather than sector fundamentals or macroeconomic conditions, may play a dominant role in driving the occurrence of bubbles.

B.4. Bubble detection results using LPPLS

See Table B.4.

B.5. Average marginal effects obtained by Logit model

See Table B.5.

Data availability

I have shared the link to my data/code at the Attach File step.

Replication Folder for "Sectoral Bubbles Formation and Contagion Dy namics in China's Stock Market" (Original data) (Mendeley Data)

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